MATH2400 Second Half

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Chapter 1

Darboux Integration

1.1 Integrability

In what follows, $a, b \in \mathbb{R}$ with $a < b$, and $f : [a, b] \to \mathbb{R}$ is a bounded function.

Definition 1.1.1

A *partition* P of $[a, b]$ is a finite set of numbers $\{x_0, x_1, x_2, \ldots, x_n\}$ such that

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

Definition 1.1.2

Suppose $P = \{x_0, x_1, x_2, \ldots, x_n\}$ is a partition of $[a, b]$. Then for $1 \leq i \leq n$,

$$m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}$$

$$M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\}$$

*The **lower (Darboux) sum** of f with respect to P, denoted by $L(f, P)$, is defined as*

$$L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}).$$

*The **upper (Darboux) sum** of f with respect to P, denoted by $U(f, P)$, is defined as*

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

Lemma 1.1.3

Let $m, M \in \mathbb{R}$ be such that for all $x \in [a, b]$, $m \leq f(x) \leq M$. Then for every partition P of $[a, b]$,

$$m(b - a) \leq L(f, P) \leq U(f, P) \leq M(b - a).$$

Proof. Let $P = \{x_0, \ldots, x_n\}$ is a partition of $[a, b]$. Note, $\forall i, m \leq m_i \leq M_i \leq M$. Also, $\sum_{i=1}^{n}(x_i - x_{i-1}) = x_n - x_0 = b - a$.

Hence,

$$m(b - a) = m\sum_{i=1}^{n}(x_i - x_{i-1}) = \sum_{i=1}^{n} m(x_i - x_{i-1})$$

$$\leq \sum_{i=1}^{n} m_i(x_i - x_{i-1}) \leq \sum_{i=1}^{n} M_i(x_i - x_{i-1}) \leq \sum_{i=1}^{n} M(x_i - x_{i-1}) = M(b - a)$$

□

Example.

Suppose $f : [0,4] \to \mathbb{R}$ be such that

$$f(x) = \begin{cases} x, & x \in [0,1) \\ 2, & x \in [1,3) \\ 5, & x \in [3,4]. \end{cases}$$

Take $P_1 = \{0,2,4\}$. Find $L(f,P_1)$ and $U(f,P_1)$.

$$L = \sum_{i=1}^{2} m_i(x_i - x_{i-1}) = 0 \cdot 2 + 2 \cdot 2 = 4$$

$$U = \sum_{i=1}^{2} M_i(x_i - x_{i-1}) = 2 \cdot 2 + 5 \cdot 2 = 14$$

Example.

Take $P_1 = \{0,1,2,3,4\}$. Find $L(f,P_2)$ and $U(f,P_2)$.

$$L = \sum_{i=1}^{4} m_i(x_i - x_{i-1}) = 0 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 + 5 \cdot 1 = 9$$

$$U = \sum_{i=1}^{4} M_i(x_i - x_{i-1}) = 2 \cdot 1 + 2 \cdot 1 + 5 \cdot 1 + 5 \cdot 1 = 14$$

Definition 1.1.4

*The partition P' is a **refinement** of P if $P \subset P'$.*

Lemma 1.1.5

If $P \subset P'$, then $L(f,P) \leq L(f,P')$ and $U(f,P) \geq U(f,P')$.

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ and first suppose $P' = \{x_0, x_1, \ldots, x_{j-1}, \gamma, x_j, \ldots, x_n\}$ (So $P \subset P'$). As before, set $m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\} \forall i$. Then

$$L(f,P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) = \sum_{i=1}^{j-1} m_i(x_i - x_{i-1}) + m_j(x_j - x_{j-1}) + \sum_{i=j+1}^{n} m_i(x_i - x_{i-1})$$

Let $m' = \inf\{f(x) : x_{j-1} \leq x \leq \gamma\}$ and $m'' = \inf\{f(x) : \gamma \leq x \leq x_j\}$. Since

$$\{f(x) : x_{j-1} \leq x \leq \gamma\} \subset \{f(x) : x_{j-1} \leq x \leq x_j\}$$

$$\Rightarrow m_j \leq m', m''$$

Proceeding,

$$L(f,P) = \sum_{i=1, i \neq j}^{n} m_i(x_i - x_{i-1}) + m_j(x_j - x_{j-1})$$

$$= \sum_{i=1, i \neq j}^{n} m_i(x_i - x_{i-1}) + m_j(\gamma - x_{j-1}) + m_j(x_j - \gamma)$$

$$\leq \sum_{i=1, i\neq j}^{n} m_i(x_i - x_{i-1}) + m'(\gamma - x_{j-1}) + m''(x_j - \gamma) = L(f, P')$$

Similarly, $U(f, P) \geq U(f, P')$. If P' contains k more points than P, repeat the argument k times. □

From Lemma 6, the sets of lower and upper Darboux sums are bounded.

Definition 1.1.6

The number

$$\underline{\int_a^b} f\,dx = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$$

is called the ***lower (Darboux) integral*** *of f.*

$$\overline{\int_a^b} f\,dx = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$$

is called the ***upper (Darboux) integral*** *of f.*

Lemma 1.1.7

If P_1, P_2 are partitions of $[a, b]$, then

$$L(f, P_1) \leq U(f, P_2).$$

Proof. Define $P = P_1 \cup P_2$. Then P is a refinement of P_1 and P_2 (or $P = P_1$ or P_2). From lemma above,

$$L(f, P_1) \leq L(f, P) \leq U(f, P) \leq U(f, P_2)$$

□

Lemma 1.1.8

(The lower integral of f is less than or equal to the upper integral of f).

$$\underline{\int_a^b} f\,dx \leq \overline{\int_a^b} f\,dx$$

Proof. We'll adopt the shorthand:

$$\mathrm{U} = \overline{\int_a^b} f\,dx = \inf\{U(f, P) : P \text{ is a partition of } [a, b]\}$$

$$\mathrm{L} = \underline{\int_a^b} f\,dx = \sup\{L(f, P) : P \text{ is a partition of } [a, b]\}$$

We wish to prove that for all f, $\mathrm{L} \leq \mathrm{U}$. Aiming for contradiction, assume that for some f, $L > U \Rightarrow \epsilon = \frac{L-U}{2} > 0$. There exists partition P_1 such that

$$L(f, P_1) > L - \epsilon = L - \frac{L - U}{2} = \frac{L + U}{2}$$

There exists partition P_2 such that

$$U(f, P_2) < U + \epsilon = U + \frac{L - U}{2} = \frac{L + U}{2}$$

Thus,

$$L(f, P_1) > \frac{U + L}{2} = U(f, P_2)$$

Contradiction. Hence, $U \geq L$. □

Definition 1.1.9

If

$$\underline{\int_a^b} f\,dx = \overline{\int_a^b} f\,dx,$$

*then f is called **integrable**, and $\int_a^b f\,dx$ is defined as*

$$\int_a^b f\,dx \equiv \underline{\int_a^b} f\,dx = \overline{\int_a^b} f\,dx.$$

***Note**, To avoid worrying about the variable of integration, we often simply write $\int_a^b f$.*

Example.

> *Consider $f : [0,1] \to \mathbb{R}$ such that $f(x) = c$*
> *Is f integrable?*

Proof. *Note that for all $x \in [0,1]$, $c \leq f(x) \leq c$*
Take any partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$ of $[0,1]$.
Then, by lemma

$$c(1-0) = L(f,P) \leq \underline{\int_a^b} f \leq \overline{\int_a^b} f \leq U(f,P) \leq c(1,0)$$

Hence, f is integrable and $\underline{\int_a}^b f = \overline{\int_a^b} f = c = \int_a^b f$ □

Theorem 1.1.10

The function $f : [a,b] \to \mathbb{R}$ is integrable if and only if for every $\epsilon > 0$, there exists a partition P such that

$$U(f,P) - L(f,P) < \epsilon.$$

Proof. $\Leftarrow$: Let $\alpha = \overline{\int_a^b} f\,dx - \underline{\int_a^b} f\,dx$. By Lemma , $\alpha \geq 0$. Suppose $\alpha > 0$. Then, by assumption, there $\exists P$ such that $U(f,P) - L(f,P) < \alpha$. Then,

$$\overline{\int_a^b} f \leq U(f,P) \quad 1)$$

$$\underline{\int_a^b} f \geq L(f,P) \quad 2)$$

Multiply 2) by -1,

$$\underline{\int_a^b} -f \leq -L(f,P) \quad 3)$$

Then 1) + 3),

$$\alpha = \overline{\int_a^b} f - \underline{\int_a^b} f \leq U(f,P) - L(f,P) < \alpha$$

Contradiction since $\alpha > 0$. Then, $\alpha = 0$ and $\overline{\int_a^b} f = \underline{\int_a^b} f = \int_a^b f$
$\Rightarrow$: $\forall \epsilon > 0$, $\exists P_1$ such that

$$\underline{\int_a^b} f - \frac{\epsilon}{2} < L(f,P_1)$$

and $\exists P_2$ such that

$$\overline{\int_a^b} f + \frac{\epsilon}{2} > U(f, P_2)$$

Let $Q = P_1 \cup P_2$. Then by lemma,

$$L(f, P_1) \leq L(f, Q) \leq U(f, Q) \leq U(f, P_2)$$

Now, suppose f is integrable, then

$$\overline{\int_a^b} f = \underline{\int_a^b} f = \int_a^b f$$

But this seems

$$\underline{\int_a^b} f - \frac{\epsilon}{2} < L(f, P_1) \leq L(f, Q)$$

$$U(f, Q) \leq U(f, P_2) < \overline{\int_a^b} f + \frac{\epsilon}{2}$$

$$\Rightarrow U(f, Q) - \frac{\epsilon}{2} < \int_a^b f < L(f, Q) + \frac{\epsilon}{2}$$

$$\Rightarrow U(f, Q) - L(f, Q) < \epsilon$$

□

Example.

> *Consider $f : [0, 1] \to \mathbb{R}$ such that*
>
> $$f = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$
>
> *Is f integrable?*

Proof. Then for all $1 \leq i \leq n$,

$$m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\} = 0$$

$$M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\} = 1$$

This means

$$L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) = 0$$

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = \sum_{i=1}^{n} (x_i - x_{i-1}) = 1$$

Take $\varepsilon = \frac{1}{2}$. Then for all P,

$$U(f, P) - L(f, P) = 1 - 0 = 1 > \varepsilon$$

$$f \text{ is not integrable.}$$

□

Example.

> *Consider $f : [0, 2] \to \mathbb{R}$ such that*
>
> $$f(x) = \begin{cases} 1, & \text{if } x < 1 \\ \frac{1}{2}, & \text{if } x = 1 \\ 0, & \text{if } x > 1 \end{cases}$$
>
> *Is f integrable?*

Proof. Let $0 < \varepsilon < 1$ be arbitrary. Consider the partition

$$P = \left\{0, 1 - \frac{\varepsilon}{4}, 1 + \frac{\varepsilon}{4}, 2\right\}.$$

Then

$$m_1 = \inf\left\{f(x) : 0 \le x \le 1 - \frac{\varepsilon}{4}\right\} = 1$$

$$m_2 = \inf\left\{f(x) : 1 - \frac{\varepsilon}{4} \le x \le 1 + \frac{\varepsilon}{4}\right\} = 0$$

$$m_3 = \inf\left\{f(x) : 1 + \frac{\varepsilon}{4} \le x \le 2\right\} = 0$$

$$\Rightarrow L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) = 1\left(1 - \frac{\varepsilon}{4}\right) + 0\left(1 + \frac{\varepsilon}{4} - \left(1 - \frac{\varepsilon}{4}\right)\right) + 0\left(2 - \left(1 + \frac{\varepsilon}{4}\right)\right) = 1 - \frac{\varepsilon}{4}$$

Also,

$$M_1 = \sup\left\{f(x) : 0 \le x \le 1 - \frac{\varepsilon}{4}\right\} = 1$$

$$M_2 = \sup\left\{f(x) : 1 - \frac{\varepsilon}{4} \le x \le 1 + \frac{\varepsilon}{4}\right\} = 1$$

$$M_3 = \sup\left\{f(x) : 1 + \frac{\varepsilon}{4} \le x \le 2\right\} = 0$$

$$\Rightarrow U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = 1\left(1 - \frac{\varepsilon}{4}\right) + 1\left(1 + \frac{\varepsilon}{4} - \left(1 - \frac{\varepsilon}{4}\right)\right) + 0\left(2 - \left(1 + \frac{\varepsilon}{4}\right)\right) = 1 + \frac{\varepsilon}{4}$$

$$\text{Hence,} \quad U(f, P) - L(f, P) = \left(1 + \frac{\varepsilon}{4}\right) - \left(1 - \frac{\varepsilon}{4}\right) = \frac{\varepsilon}{2} < \varepsilon.$$

So f is integrable.

Furthermore,

$$1 - \frac{\varepsilon}{4} = L(f, P) \le \int_0^2 f\,dx \le U(f, P) = 1 + \frac{\varepsilon}{4}$$

$$\Longrightarrow \left|\int_0^2 f\,dx - 1\right| \le \frac{\varepsilon}{4}.$$

Since ε was arbitrary, we conclude

$$\int_0^2 f\,dx = 1.$$

□

Theorem 1.1.11

If $f : [a, b] \to \mathbb{R}$ is continuous at all but finitely many points, then f is integrable.

Proof. We will prove this for the simpler case that f is continuous throughout $[a, b]$. The more general case is 5.2.8 in Lebl.

As f is continuous on a closed bounded interval, it is uniformly continuous. Let $\epsilon > 0$ be given. Then there exists $\delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\epsilon}{b-a}$.

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ such that $\Delta x_i < \delta$ for all $i = 1, 2, \ldots, n$. For example, take n such that $\frac{b-a}{n} < \delta$ and let $x_i = \frac{i(b-a)}{n} + a$. Then for all $x_i, y_i \in [x_{i-1}, x_i]$,

$$|x_i - y_i| \le \Delta x_i < \delta \Rightarrow |f(x_i) - f(y_i)| < \frac{\epsilon}{b - a}.$$

As f is continuous on $[x_{i-1}, x_i]$, it attains a maximum and minimum. Let x_i (y_i) be the point where f attains M_i (resp. m_i). Thus

$$f(x_i) = M_i, \quad f(y_i) = m_i$$

Then,

$$|x_i - y_i| \leq \Delta x_i < \delta \Rightarrow f(x_i) - f(y_i) < |f(x_i) - f(y_i)| < \frac{\epsilon}{b-a}.$$

$$\begin{aligned} U(f,P) - L(f,P) &= \left(\sum_{i=1}^{n} M_i \Delta x_i\right) - \left(\sum_{i=1}^{n} m_i \Delta x_i\right) \\ &= \sum_{i=1}^{n} (M_i - m_i)\Delta x_i \\ &= \sum_{i=1}^{n} \left(f(x_i) - f(y_i)\right) \Delta x_i \\ &< \frac{\epsilon}{b-a} \sum_{i=1}^{n} \Delta x_i \\ &= \frac{\epsilon}{b-a}(b-a) = \epsilon \end{aligned}$$

As $\epsilon > 0$ was arbitrary, the result follows from Thm 10.

□

Theorem 1.1.12

If $f : [a,b] \to \mathbb{R}$ is monotone (i.e., non-increasing or non-decreasing), then f is integrable.

Proof. A monotone function on $[a,b]$ is necessarily bounded.

Assume f is non-decreasing (the non-increasing case is similar).

Fix $\epsilon > 0$. We want to find a partition $P = \{x_0, x_1, x_2, \ldots, x_N\}$ of $[a,b]$ such that

$$U(f,P) - L(f,P) < \epsilon.$$

Assume P splits into equal parts.

Thus,

$$x_k = a + \frac{b-a}{N}k, \quad k = 0, 1, \ldots, N$$

$$m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\} = f(x_{i-1})$$

$$M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\} = f(x_i)$$

Then,

$$L(f,P) = \sum_{i=1}^{N} f(x_{i-1})(x_i - x_{i-1})$$

$$U(f,P) = \sum_{i=1}^{N} f(x_i)(x_i - x_{i-1})$$

$$\Rightarrow U(f,P) - L(f,P) = \sum_{i=1}^{N} (f(x_i) - f(x_{i-1}))\frac{b-a}{N}$$

$$= \frac{b-a}{N}\left(f(x_1) - f(x_0) + f(x_2) - f(x_1) + \ldots + f(x_N) - f(x_{N-1})\right)$$

$$= \frac{b-a}{N}(f(b) - f(a))$$

Choose

$$N = \left\lceil \frac{(b-a)(f(b) - f(a))}{\epsilon} \right\rceil + 1$$

$$\Rightarrow \frac{(b-a)(f(b) - f(a))}{N} < \epsilon$$

$$\therefore \quad \frac{(b-a)(f(b)-f(a))}{N} < \epsilon$$

Thus, f is integrable. □

One interprets the Darboux integral as the signed area under the graph.

1.2 Properties of the Integral

Theorem 1.2.1

If f, g are integrable on $[a, b]$, then the following apply:

1. If $k \in \mathbb{R}$, then kf is integrable, and
$$\int_a^b kf\,dx = k\int_a^b f\,dx.$$
2. $f + g$ is integrable, and
$$\int_a^b (f+g)\,dx = \int_a^b f\,dx + \int_a^b g\,dx.$$
3. If $f \leq g$ on $[a, b]$, then
$$\int_a^b f\,dx \leq \int_a^b g\,dx.$$
4. If f is integrable on both $[a, b]$ and $[b, c]$, then f is integrable on $[a, c]$, and
$$\int_a^c f\,dx = \int_a^b f\,dx + \int_b^c f\,dx.$$
5. For every $d \in [a, b]$, f is integrable on $[a, d]$ and $[d, b]$.
6. $|f|$ is integrable, and
$$\left|\int_a^b f\,dx\right| \leq \int_a^b |f|\,dx.$$
(Triangle inequality for integrals).

Definition 1.2.2

Let $a < b$, and suppose f is integrable on $[a, b]$. Then

$$\int_a^a f\,dx \equiv 0$$

$$\int_b^a f\,dx \equiv -\int_a^b f\,dx.$$

Theorem 1.2.3: Mean value theorem for integrals

If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ such that

$$\int_a^b f\,dx = f(c)(b-a).$$

Proof. Since f is continuous, it is integrable (Thm 11).

By the EVT (Extreme Value Theorem), $\exists\, x_m, x_M$ such that

$$f(x_m) = \min_{[a,b]} f(x), \quad f(x_M) = \max_{[a,b]} f(x)$$

Then, from Theorems 6 and 9,

$$f(x_m)(b-a) \leq \int_a^b f \leq f(x_M)(b-a)$$

$$f(x_m) \leq \frac{1}{b-a}\int_a^b f \leq f(x_M)$$

By the IVT (Intermediate Value Theorem), $\exists\, c \in (a,b)$ such that

$$\frac{1}{b-a}\int_a^b f = f(c)$$

□

Definition 1.2.4

F is called an *anti-derivative* of f if $F' = f$.

Theorem 1.2.5: Fundamental Theorem of Calculus

Suppose $f : [a,b] \to \mathbb{R}$ is continuous.
Define

$$F(x) = \int_a^x f(t)\,dt.$$

Then F is an antiderivative of f, and

$$\int_a^b f(t)\,dt = F(b) - F(a).$$

Proof. F is well defined because f is continuous and hence integrable on $[a,x]$ for each $x \in [a,b]$.

We claim that $F' = f$ on (a,b). Take $x \in (a,b)$.

We will show that

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h} = f(x).$$

We first fix $h \in (a-x, b-x) \setminus \{0\}$. Then $F(x+h)$ is defined and

$$\begin{aligned} F(x+h) - F(x) &= \int_a^{x+h} f - \int_a^x f \\ &= \int_x^{x+h} f \\ &= \frac{h}{|h|}\int_{\min(x,x+h)}^{\max(x,x+h)} f \\ &= \frac{h}{|h|} f(c_h)|h| \\ &= hf(c_h), \end{aligned}$$

where $\min(x, x+h) \leq c_h \leq \max(x, x+h)$

by the Mean Value Theorem for Integrals.

$$\frac{F(x+h) - F(x)}{h} = f(c_h)$$

Now define a function $c : (a-x, b-x) \setminus \{0\} \to \mathbb{R}$ such that $c(h) = c_h$. Note,

$$c(h) \text{ is between } x \text{ and } x+h.$$

$$\text{Notice that } \lim_{h\to 0} x = \lim_{h\to 0} x+h = x$$

$$\text{Using the squeeze theorem, then } \lim_{h\to 0} c(h) = x.$$

$$\text{Since } f \text{ is continuous at } x, \text{ we have}$$

$$\lim_{h\to 0} \frac{F(x+h)-F(x)}{h} = \lim_{h\to 0} f(c(h)) = f\left(\lim_{h\to 0} c(h)\right) = f(x) \text{ as required.}$$

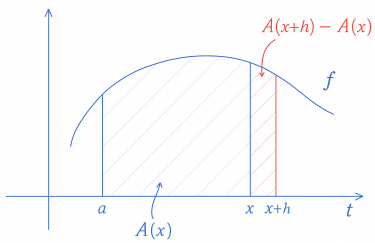
To prove the second part, note that:

$$F(a) = \int_a^a f = 0,$$

$$F(b) = \int_a^b f,$$

$$\Rightarrow \int_a^b f = F(b) - F(a) = F(b) - 0 = F(b).$$

□



Theorem 1.2.6

If $G' = f$ on (a,b), then for all x, $\int_a^b f(t)\,dt = G(b) - G(a)$.

Proof.

$$(G-F)'(x) = G'(x) - F'(x) = f(x) - f(x) = 0$$

Therefore,

$$G - F = C \in \mathbb{R} \Rightarrow G = F + C$$

$$G(b) - G(a) = (G(b) - C) - (G(a) - C) = F(b) - F(a) = \int_a^b f(t)\,dt$$

□

If f is discontinuous anywhere, $F(x) = \int_a^x f(t)\,dt$ might not be differentiable.

Example.

Take $f(x) = sign(x)$ on $[-1,1]$.
Then

$$f(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

$$\int_{-1}^x f(t)\,dt = |x| - 1 \quad \text{on } [-1,1].$$

$$(|x|-1)' \quad \text{DNE when } x=0.$$

Theorem 1.2.7: Integration by parts

Suppose that functions u, v, u', v' are all continuous on $[a,b]$. Then

$$\int_a^b uv'\,dx = uv\Big|_a^b - \int_a^b vu'\,dx$$

Proof. The product rule yields

$$(uv)' = u'v + uv'.$$

All functions are continuous. Thus, $u'v$ and uv' are continuous, and therefore so is $(uv)'$.

Hence, all three are integrable from a to b, and

$$\int_a^b (uv)'\,dx = \int_a^b u'v\,dx + \int_a^b uv'\,dx.$$

$$\Rightarrow uv\Big|_a^b = \int_a^b u'v\,dx + \int_a^b uv'\,dx.$$

(From the Fundamental Theorem of Calculus and the fact that the integral of the sum is the sum of the integrals.)

Re-arranging, we have our result. □

1.3 Improper Integrals

Definition 1.3.1

Consider a function $f : (a,b] \to \mathbb{R}$, not necessarily bounded.
Assume f is integrable on $[a+\epsilon, b]$ for every $\epsilon \in (0, b-a)$.
Then

$$\lim_{\epsilon\to 0^+} \int_{a+\epsilon}^b f\,dx \text{ is called the improper integral (of the first kind) of } f \text{ on } [a,b].$$

We usually denote this simply by

$$\int_a^b f\,dx.$$

Definition 1.3.2

Assume $f : [a,\infty) \to \mathbb{R}$ is integrable on $[a,b]$ for every $b > a$. Then

$$\int_a^\infty f\,dx = \lim_{b\to\infty} \int_a^b f\,dx$$

is called the improper integral (of the second kind) of f on $[a,\infty)$.
We define $\int_{-\infty}^b f\,dx$ similarly.

Definition 1.3.3

$$\int_{-\infty}^\infty f\,dx = \int_{-\infty}^0 f\,dx + \int_0^\infty f\,dx$$

provided both integrals on the right exist.

$$\int_{-\infty}^{\infty} f\,dx \neq \lim_{a\to\infty}\int_{-a}^{a} f\,dx.$$

Example.

$$\int_{-\infty}^{\infty} x\,dx$$

Proof.

$$\int_0^{\infty} x\,dx = \lim_{b\to\infty}\int_0^b x\,dx = \lim_{b\to\infty}\left.\frac{x^2}{2}\right|_0^b = \lim_{b\to\infty}\frac{b^2}{2}$$

DNE.

However,

$$\begin{aligned}\lim_{a\to\infty}\int_{-a}^{a} x\,dx &= \lim_{a\to\infty}\left.\frac{x^2}{2}\right|_{-a}^{a}\\ &= \lim_{a\to\infty}\left(\frac{a^2}{2}-\frac{(-a)^2}{2}\right)\\ &= \lim_{a\to\infty} 0 = 0\end{aligned}$$

□

Example.

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{\epsilon\to 0^+}\int_\epsilon^1 \frac{1}{\sqrt{x}}\,dx$$

Proof.

$$= \lim_{\epsilon\to 0^+} 2\sqrt{x}\Big|_\epsilon^1 = 2\lim_{\epsilon\to 0^+}\left(1-\sqrt{\epsilon}\right) = 2$$

□

Example.

More generally, consider

$$\int_0^1 x^p\,dx.$$

(a) $p > 0$

(b) $p \in (-1, 0]$

(c) $p < -1$

(d) $p = -1$

(a) x^p is continuous on $[0, 1]$, so it is integrable.

(b)

$$\int_0^1 x^p\,dx = \lim_{\epsilon\to 0^+}\int_\epsilon^1 x^p\,dx = \lim_{\epsilon\to 0^+}\left.\frac{x^{p+1}}{p+1}\right|_\epsilon^1$$

$$= \frac{1}{p+1}\lim_{\epsilon\to 0^+}\left(1-\epsilon^{p+1}\right) = \frac{1}{p+1}.$$

(c) Same argument shows that the integral diverges.

$$\epsilon^{p+1} = \frac{1}{\epsilon^{-(p+1)}}$$

(d)

$$\int_0^1 \frac{1}{x}\,dx = \lim_{\epsilon\to 0^+} \int_\epsilon^1 \frac{1}{x}\,dx = \lim_{\epsilon\to 0^+} \ln x\big|_\epsilon^1 = -\lim_{\epsilon\to 0^+} \ln \epsilon$$

We will show this limit does not exist.

Theorem 1.3.4: Integral test

Assume f is continuous, non-negative, and nonincreasing on $[1,\infty)$. Then

$$\int_1^x f\,dx, \quad \text{and} \quad \sum_{n=1}^{\infty} f(n)$$

converge or diverge simultaneously.

Proof. Consider the partition $\{1, 2, \ldots, n, n+1\}$ of the interval $[1, n+1]$. Then because f is non-increasing,

$$\inf\{f(x) : x_{i-1} \le x \le x_i\} = f(x_i)$$
$$\sup\{f(x) : x_{i-1} \le x \le x_i\} = f(x_{i-1})$$

where $x_0 = 1, x_1 = 2, \ldots$. Therefore,

$$\begin{aligned} U(f,P) &= \sum_{i=1}^{n} f(x_{i-1})(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} f(x_{i-1})(i+1-i) \\ &= \sum_{i=1}^{n} f(x_{i-1}) \\ &= \sum_{i=1}^{n} f(i). \end{aligned}$$

Similarly,

$$\begin{aligned} L(f,P) &= \sum_{i=1}^{n} f(x_i)(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} f(i+1) \\ &= \sum_{k=2}^{n+1} f(k). \end{aligned}$$

Now for all P,

$$L(f,P) \le \int_1^{n+1} f(x)\,dx \le U(f,P)$$

From the first inequality, we have

$$\sum_{k=2}^{n+1} f(k) \le \int_1^{n+1} f(x)\,dx \Rightarrow \sum_{k=2}^{n} f(k) \le \int_1^{n} f(x)\,dx \Rightarrow \sum_{k=1}^{n} f(k) \le f(1) + \int_1^{n} f(x)\,dx.$$

Suppose $\int_1^\infty f(x)\,dx$ is convergent.

Then it has a finite value, and because f is non-negative, we know that

$$\sum_{k=1}^{\infty} f(k) \leq f(1) + \int_1^{\infty} f(x)\,dx = M.$$

Since $\sum_{k=1}^{n} f(k)$ is monotonically increasing and bounded by M, it converges.

From the second inequality,

$$\int_1^n f(x)\,dx \leq \sum_{i=1}^{n-1} f(i) \leq \sum_{i=1}^{n} f(i)$$

Since $\int_1^n f(x)\,dx$ is monotonically increasing with n, it is bounded above if and only if it converges.

Suppose $\int_1^{\infty} f(x)\,dx$ diverges,
then it is unbounded, and $\sum_{i=1}^{n} f(i)$ also diverges as $n \to \infty$

□

From the last few lectures, we can now construct an alternate definition of ln and exp:

Definition 1.3.5

Define $\ln : (0, \infty) \to \mathbb{R}$ *such that*

$$\ln(x) = \int_1^x \frac{1}{t}\,dt.$$

Lemma 1.3.6

$\ln : (0, \infty) \to \mathbb{R}$ is invertible

Proof. Since

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges (harmonic series, see standard analysis references), $\ln x$ has no upper bound.

Thus,

$$\forall M > 0, \exists d : \ln(d) > M.$$

Since

$$(\ln x)' = \frac{1}{x},$$

it follows that ln is differentiable, and thus continuous.

Since $\ln 1 = 0$, there exists c such that $1 < c < d$ and

$$\ln(c) = M$$

(by I.V.T.).

Also,

$$\ln(c^{-1}) = -\ln(c) = -M,$$

thus ln is onto.

Since

$$(\ln x)' = \frac{1}{x} > 0 \quad \forall x > 0,$$

ln is strictly increasing, which implies ln is one-to-one.

Thus, ln is invertible. □

Definition 1.3.7

Define $\exp : \mathbb{R} \to (0, \infty)$ *such that*

$$\exp(\ln(x)) = \ln(\exp(x)) = x.$$

That is *exp* is the inverse of *ln*

From the inverse function theorem, setting $y = \ln x$,

$$(\exp y)' = \frac{1}{(\ln(\exp y))'} = \frac{1}{\frac{1}{\exp y}} = \exp(y)$$

All other properties of ln and exp can also be proven. Many other functions can be constructed this way.

Chapter 2

Sequences and Series of Functions

2.1 Sequences of Functions

2.1.1 Pointwise and uniform convergence

Let $\Omega \subseteq \mathbb{R}$ and suppose $f_n : \Omega \to \mathbb{R}$ is a function for each $n \in \mathbb{N}$. We consider the sequence $\{f_n\} \equiv \{f_n\}_{n=1}^{\infty}$.

Example.

$$f_1(x) = x, \quad f_2(x) = x^2, \quad f_3(x) = x^3, \ldots \quad \text{on } [2, 34].$$

$$f_n(x) = x^n$$

Example.

$$f_1(x) = x, \quad f_2(x) = 2x^2, \quad f_3(x) = 3x^3, \ldots \quad \text{on } [-1, \pi].$$

$$f_n(x) = nx^n$$

Example.

$$f_n(x) = \frac{x}{n} \quad \text{on } [0, 1].$$

Definition 2.1.1

*The sequence $\{f_n\}$ **converges pointwise** to $f : \Omega \to \mathbb{R}$ if*

$$\lim_{n\to\infty} f_n(x) = f(x) \text{ for each } x \in \Omega.$$

*The sequence $\{f_n\}$ converges **pointwise** to f if for all $x \in \Omega$, and for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (possibly depending on ε and x) such that for all $n \geq N$,*

$$|f_n(x) - f(x)| < \varepsilon.$$

Note: $N = N(\varepsilon, x)$.

Example.

$f_n(x) = x^n$ on $[0, 1]$ *Converges pointwise.*

Proof. If $x \in [0, 1)$, then

$$\lim_{n\to\infty} x^n = 0.$$

However,

$$\lim_{n\to\infty} f_n(1) = \lim_{n\to\infty} 1^n = 1.$$

Thus, f_n converges pointwise to

$$f(x) = \begin{cases} 0 & \text{if } x \in [0,1), \\ 1 & \text{if } x = 1. \end{cases}$$

□

Even though f_n is continuous for all $n \in \mathbb{N}$, the pointwise limit f is discontinuous. This motivates a different definition of convergence.

Definition 2.1.2

Let $f : \Omega \to \mathbb{R}$. The sequence $\{f_n\}$ converges ***uniformly*** *to f (written $f_n \xrightarrow[n\to\infty]{} f$) if: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that if $n \geq N$, then*

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in \Omega.$$

Note: $N = \mathcal{N}(\varepsilon)$

Theorem 2.1.3

If $f_n : \Omega \to \mathbb{R}$ is continuous for every $n \in \mathbb{N}$, and if f_n converges uniformly to $f : \Omega \to \mathbb{R}$, then f is continuous.

Proof. Fix $\varepsilon > 0$, and let $x \in \Omega$.

1. Since f_N is continuous on Ω, it is continuous at x. Thus, there exists $\delta > 0$ such that for all $y \in \Omega$ with $|x - y| < \delta$,
$$|f_N(x) - f_N(y)| < \frac{\varepsilon}{3}.$$
2. Since $f_n \xrightarrow[n\to\infty]{} f$, there exists $N \in \mathbb{N}$ such that
$$|f_n(t) - f(t)| < \frac{\varepsilon}{3} \quad \text{for all } t \in \Omega \text{ and all } n \geq N.$$

With these two facts in hand, suppose $y \in \Omega$ and $|x - y| < \delta$. Then,

$$\begin{aligned} |f(x) - f(y)| &= |f(x) - f_N(x) + f_N(x) - f_N(y) + f_N(y) - f(y)| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| \\ &< \underbrace{\frac{\varepsilon}{3}}_{\text{U.C.}} + \underbrace{\frac{\varepsilon}{3}}_{\text{cts at } x} + \underbrace{\frac{\varepsilon}{3}}_{\text{U.C.}} = \varepsilon. \end{aligned}$$

Since ε was arbitrary, f is continuous at x.

Since x was chosen from Ω arbitrarily, f is continuous on Ω.

□

Uniform convergence implies pointwise convergence, but not the other way around!

Example.

> *Let $f_n(x) = x^n$ on $[0,1]$.* We know that $\{f_n\}$ converges pointwise to
>
> $$f(x) = \begin{cases} 0, & x \in [0,1) \\ 1, & x = 1. \end{cases}$$
>
> Let us show that it **does not** converge uniformly.

The easy way to show this is to note that f_n is continuous, but f is not. Hence, $\{f_n\}$ cannot converge uniformly.

Proof. We can also show this using the definition: We will show that there exists $\varepsilon > 0$ such that for any $n \in \mathbb{N}$, there exists $x \in [0,1]$ with

$$|f_n(x) - f(x)| \geq \varepsilon.$$

Crucially, x can vary for different values of n.

Take $\varepsilon = \frac{1}{2}$, and take $x = \left(\frac{3}{4}\right)^{1/n}$. Then, for all n,

$$0 \leq \left(\frac{3}{4}\right)^{1/n} < 1,$$

so

$$f\left(\left(\frac{3}{4}\right)^{1/n}\right) = 0.$$

Also,

$$f_n\left(\left(\frac{3}{4}\right)^{1/n}\right) = \left(\left(\frac{3}{4}\right)^{1/n}\right)^n = \frac{3}{4}.$$

We then have:

$$\left|f_n\left(\left(\frac{3}{4}\right)^{1/n}\right) - f\left(\left(\frac{3}{4}\right)^{1/n}\right)\right| = \left|\frac{3}{4} - 0\right| = \frac{3}{4} \geq \frac{1}{2}.$$

□

Note: $\lim_{n\to\infty} \left(\frac{3}{4}\right)^{1/n} = 1$. So our counterexample involved taking values of f closer and closer to f's point of discontinuity.

Example.

> *However, using the same mapping as the previous example,*
>
> $$f_n \underset{n\to\infty}{\Longrightarrow} f \quad \text{on} \quad [0,p] \quad \text{for every} \quad p < 1.$$

Proof. Fix $\varepsilon > 0$. We need to find $N \in \mathbb{N}$ such that if $n \geq N$, then

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in [0,p].$$

Note: N can depend on ε, but it must work for *all* $x \in [0,p]$.

Now, $f(x) = 0$ on $[0,p]$, and $f_n(x) = x^n$. Thus, we need to show there exists $N = N(\varepsilon)$ such that for all $n \geq N$,

$$|x^n - 0| = x^n < \varepsilon \quad \text{for all } x \in [0,p].$$

Now, since $x^n = x^N x^{n-N} \leq x^N \leq p^N$, if we can find $N = N(\varepsilon)$ such that

$$p^N < \varepsilon, \quad (*)$$

we are done.

Since ln is a strictly increasing function, (*) is true if and only if

$$\ln(p^N) < \ln(\varepsilon) \quad \Longleftrightarrow \quad N \ln p < \ln \varepsilon \quad \Longleftrightarrow \quad N > \frac{\ln \varepsilon}{\ln p}.$$

Set

$$N = \left\lceil \frac{\ln \varepsilon}{\ln p} \right\rceil + 1.$$

The result follows.

□

Example.

Let $f_n(x) = \sin^n(x)$, $x \in \mathbb{R}$. Is this uniformly convergent?

Proof. No! Not even pointwise convergent on any interval which includes the point

$$2m\pi + \frac{3\pi}{2}, \quad m \in \mathbb{N}.$$

This is because

$$\sin^n\left(2m\pi + \frac{3\pi}{2}\right) = (-1)^n,$$

which does not converge. □

Example.

Is it pointwise convergent on any sub-intervals of $\mathbb{R}$? Is it uniformly convergent on any of these?

2.1.2 Interchange of limits

Expressed using limits, theorem concerned about the conditions on f_n such that

$$\lim_{x\to a}\left(\lim_{n\to\infty} f_n(x)\right) = \lim_{n\to\infty}\left(\lim_{x\to a} f_n(x)\right).$$

It often isn't possible to interchange limit operations.

Example.

Is

$$\lim_{n\to\infty}\left(\lim_{k\to\infty}\frac{n}{n+k}\right) = \lim_{k\to\infty}\left(\lim_{n\to\infty}\frac{n}{n+k}\right)?$$

Proof. Let's evaluate both sides.

LHS:

$$= \lim_{n\to\infty}\left(\lim_{k\to\infty}\frac{n}{n+k}\right) = \lim_{n\to\infty} 0 = 0$$

RHS:

$$\lim_{k\to\infty}\left(\lim_{n\to\infty}\frac{n}{n+k}\right) = \lim_{k\to\infty} 1 = 1$$

Conclusion: No! These two limits are not equal. □

What about interchanging other kinds of limit operations?

Example.

Let

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \quad \text{for} \quad x \in \mathbb{R}.$$

Is

$$\lim_{n\to\infty}\left(\frac{d}{dx}f_n(x)\right) = \frac{d}{dx}\left(\lim_{n\to\infty} f_n(x)\right)?$$

Proof. LHS:

$$\lim_{n\to\infty}\left(\sqrt{n}\cos(nx)\right)$$

does not exist for some values of x. For instance,

$$\lim_{n\to\infty}\left(\sqrt{n}\cos(n\times 0)\right) = \lim_{n\to\infty}\sqrt{n} = \infty,$$

thus DNE.

RHS:

$$= \frac{d}{dx}\left(\lim_{n\to\infty} f_n(x)\right) = \frac{d}{dx}(0) = 0.$$

for all x, Thus, the equality does not hold.

□

Example.

Let

$$f_n(x) = n^2 x(1-x^2)^n \quad \text{for} \quad x \in [0,1].$$

Is

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx \ ?$$

Proof. LHS:

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \lim_{n\to\infty} \frac{n^2}{2n+2} = \infty,$$

thus the limit does not exist.

RHS:

$$\int_0^1 \lim_{n\to\infty} f_n(x)\,dx = \int_0^1 0\,dx = 0.$$

Thus, the equality does not hold. □

For this example, we'll directly show that f_n is not uniformly convergent.

Proof. Set $x = 1/\sqrt{n}$. Then

$$f_n\left(\frac{1}{\sqrt{n}}\right) = n^2 \frac{1}{\sqrt{n}}\left(1-\frac{1}{n}\right)^n = n^{3/2}\left(1-\frac{1}{n}\right)^n.$$

Note that

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n = e^{-1}.$$

This means $\exists N \in \mathbb{R}$ such that $\forall n \geq N$,

$$\left|\left(1-\frac{1}{n}\right)^n - e^{-1}\right| < \frac{e^{-1}}{2}$$

which implies

$$-\frac{e^{-1}}{2} < \left(1-\frac{1}{n}\right)^n - e^{-1} < \frac{e^{-1}}{2}$$

thus

$$\frac{e^{-1}}{2} < \left(1-\frac{1}{n}\right)^n.$$

This means that for all $n \geq N$,

$$f_n\left(\frac{1}{\sqrt{n}}\right) > n^{3/2}\frac{e^{-1}}{2},$$

hence

$$\lim_{n\to\infty} f_n\left(\frac{1}{\sqrt{n}}\right) = \infty.$$

Thus, f_n is not uniformly convergent. □

Theorem 2.1.4

Suppose that $\{f_n\}$ is a sequence of functions which are integrable on $[a,b]$, and that $\{f_n\}$ converges uniformly on $[a,b]$ to a function f which is integrable on $[a,b]$. Then

$$\lim_{n\to\infty}\int_a^b f_n = \int_a^b f.$$

This is equivalent to saying

$$\lim_{n\to\infty}\int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n.$$

Proof. Fix $\varepsilon > 0$. There exists some $N \in \mathbb{N}$ such that $\forall n \geq N$, we have

$$|f(x) - f_n(x)| < \frac{\varepsilon}{2(b-a)} \quad \forall x \in [a,b].$$

Thus, if $n \geq N$, we have

$$\begin{aligned}\left|\int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx\right| &= \left|\int_a^b (f_n(x) - f(x))\,dx\right| \\ &\leq \int_a^b |f_n(x) - f(x)|\,dx \\ &\leq \int_a^b \frac{\varepsilon}{2(b-a)}\,dx \\ &= \frac{\varepsilon}{2(b-a)}(b-a) = \frac{\varepsilon}{2} < \varepsilon.\end{aligned}$$

□

Theorem 2.1.5

Let $\{f_n\}$ be a sequence of continuously differentiable functions defined on $[a,b]$. If

(a) *The sequence $\{f_n\}$ converges pointwise to f and*

(b) *The sequence of derivatives $\{f_n'\}$ converges uniformly,*

then f is differentiable, and

$$f' = \lim_{n\to\infty} f_n'.$$

In other words, for all x,

$$\frac{d}{dx}\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty}\frac{d}{dx} f_n(x).$$

Proof. Define

$$g = \lim_{n\to\infty} f_n'.$$

Then, since (f_n') is continuous and uniformly convergent, g is continuous.

Let $x \in [a,b]$. Since g is continuous, it is integrable, and so

$$\begin{aligned}\int_a^x g = \int_a^x \lim_{n\to\infty} f_n' &= \lim_{n\to\infty}\int_a^x f_n' \\ &= \lim_{n\to\infty}(f_n(x) - f_n(a)) \quad \text{(by FTC)} \\ &= f(x) - f(a) \quad \text{(by pointwise convergence of } f_n \text{ to } f).\end{aligned}$$

Since g is continuous, we have, by the first part of FTC,

$$g(x) = \frac{d}{dx}\left(\int_a^x g\right) = \frac{d}{dx}(f(x) - f(a)) = f'(x).$$

□

Definition 2.1.6

The sequence of functions $\{f_n\}_{n=1}^{\infty}$ (defined over Ω) is uniformly Cauchy if and only if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that if $l, m \geq N$, then

$$|f_l(x) - f_m(x)| < \epsilon \quad \text{for all} \quad x \in \Omega.$$

Theorem 2.1.7: Cauchy convergence criterion

The sequence of functions $\{f_n\}_{n=1}^{\infty}$ converges uniformly to some function f if and only if it is uniformly Cauchy.

Proof. $\Rightarrow$: Assume $f_n \to f$ uniformly. Fix $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that if $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2} \quad \text{for all} \quad x \in \Omega.$$

If $l, m \geq N$,

$$\begin{aligned} |f_l(x) - f_m(x)| &= |f_l(x) - f(x) + f(x) - f_m(x)| \\ &\leq |f_l(x) - f(x)| + |f_m(x) - f(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \quad \text{for all} \quad x \in \Omega. \end{aligned}$$

$\Leftarrow$: For all $x \in \Omega$, $\{f_n\}$ is Cauchy. Therefore, it converges at least pointwise. Define

$$f : \Omega \to \mathbb{R} \quad \text{by} \quad f(x) = \lim_{n\to\infty} f_n(x).$$

We need to prove $f_n \to f$ uniformly.

Fix $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that if $m, n \geq N$, then for all $x \in \Omega$,

$$|f_m(x) - f_n(x)| < \frac{\varepsilon}{2}.$$

Thus,

$$f_n(x) - \frac{\varepsilon}{2} < f_m(x) < f_n(x) + \frac{\varepsilon}{2}.$$

Fix n and consider the first inequality:

$$f_n(x) - \frac{\varepsilon}{2} < f_m(x).$$

Taking limits as $m \to \infty$,

$$f_n(x) - \frac{\varepsilon}{2} \leq f(x).$$

Similarly, the second inequality shows:

$$f_m(x) < f_n(x) + \frac{\varepsilon}{2}.$$

Thus, taking limits,

$$f(x) = \lim_{m\to\infty} f_m(x) \leq f_n(x) + \frac{\varepsilon}{2}.$$

Bringing these together, we have

$$f_n(x) - \frac{\varepsilon}{2} \leq f(x) \leq f_n(x) + \frac{\varepsilon}{2},$$

thus

$$|f(x) - f_n(x)| \leq \frac{\varepsilon}{2} < \varepsilon.$$

□

2.2 Series of Functions

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions from Ω to $\mathbb{R}$.

We consider

$$\sum_{n=1}^{\infty} f_n(x) \equiv \lim_{N\to\infty} \sum_{n=1}^{N} f_n(x).$$

Denote S_k such that

$$S_k(x) = f_1(x) + f_2(x) + \cdots + f_k(x) = \sum_{i=1}^{k} f_i(x).$$

Definition 2.2.1

The series $\sum_{n=1}^{\infty} f_n$ converges pointwise (respectively, uniformly) if the sequence $\{S_k\}_{k=1}^{\infty}$ converges pointwise (respectively, uniformly).

Theorem 2.2.2: Weierstrass M-test

Suppose $\sum_{n=1}^{\infty} M_n$ is a convergent series with non-negative elements. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions such that

$$|f_n(x)| \leq M_n \quad \text{for all} \quad x \in \Omega.$$

Then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on Ω.

Proof. Let $S_k(x) = \sum_{l=1}^{k} f_l(x)$, and suppose $\sum_{n=1}^{\infty} M_n$ converges. Then, by Cauchy's convergence test, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that if $n, m \geq N$, then

$$\sum_{l=m+1}^{n} M_l < \varepsilon.$$

Now,

$$|S_n(x) - S_m(x)| = \left| \sum_{l=m+1}^{n} f_l(x) \right|$$

$$\leq \sum_{l=m+1}^{n} |f_l(x)| \quad \text{(by the triangle inequality)}$$

$$\leq \sum_{l=m+1}^{n} M_l$$

$$< \varepsilon \quad \text{for all} \quad x \in \Omega.$$

Thus, by Cauchy convergence criterion for uniform convergence, the series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on Ω. □

Example.

Does

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

converge uniformly?

Proof. Note that

$$\left|\frac{\sin(nx)}{n^2}\right| \leq \frac{1}{n^2}.$$

Now, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges (by the p-test with $p = 2 > 1$).

Therefore, by Weierstrass M-test, the series

$$\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

does converge uniformly. □

Theorem 2.2.3

Let $\sum_{n=1}^{\infty} f_n$ converge at least pointwise to f on $[a, b]$.
If the convergence is uniform, the following can be said:

1. If each f_n is continuous on $[a, b]$, then f is continuous on $[a, b]$.
2. If f and each f_n is integrable on $[a, b]$, then

$$\int_a^b f = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Meanwhile, if each f_n' is continuous and $\sum_{n=1}^{\infty} f_n'$ converges uniformly on $[a, b]$ to some continuous function, then

$$f'(x) = \sum_{n=1}^{\infty} f_n'(x) \quad \text{for all} \quad x \in [a, b].$$

That is we can move the limits around.

Proof. Let

$$S_k = \sum_{n=1}^{k} f_n,$$

1. If each f_n is continuous, then so is S_k. Since $S_k \xrightarrow{k\to\infty} f$, it follows that f is continuous.
2. Since $S_k \xrightarrow{k\to\infty} f$, it follows that

$$\int_a^b f = \lim_{k\to\infty} \int_a^b S_k = \lim_{k\to\infty} \int_a^b \left(\sum_{n=1}^{k} f_n\right) = \lim_{k\to\infty} \sum_{n=1}^{k} \int_a^b f_n = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Lastly, note that $\lim_{k\to\infty} S_k = f$. Since each f_n' is continuous, so is

$$S_k' = \sum_{n=1}^{k} f_n'.$$

Since $\{S'_k\}$ converges uniformly to a continuous function, it follows that

$$f' = \lim_{k\to\infty} S'_k = \lim_{k\to\infty} \sum_{n=1}^{k} f'_n = \sum_{n=1}^{\infty} f'_n.$$

□

2.3 Taylor Series

Application: Function approximations, defining new functions, solving ODEs...

Definition 2.3.1

Let $\{c_k\}$ be a (possibly infinite) sequence.
The following is a shorthand for an n-th degree polynomial:

$$\sum_{k=0}^{n} c_k(x-a)^k \equiv c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots + c_n(x-a)^n.$$

Note: by this definition,

$$\sum_{k=0}^{0} c_k(x-a)^k = c_0.$$

I'm almost, but not quite, saying that $0^0 = 1$. Throughout the chapter, I'll be using this shorthand. This is only a shorthand though; if you don't like to use it, you can always consider cases like $n = 0$ separately. It just takes longer!

Definition 2.3.2

If f is a function defined on an open interval I, define $f^{(n)}(a)$ to be the n-th order derivative of f at the point $a \in I$.

Definition 2.3.3

Suppose f is n-times differentiable at a.
The n-th *Taylor polynomial* of f at a is

$$P_{n,a}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Definition 2.3.4

If it exists, we call

$$\lim_{n\to\infty} P_{n,a}(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

the *Taylor Series* for f (about a). In the special case where $a = 0$, we call this the *Maclaurin Series* of f.

Theorem 2.3.5: Taylor's theorem - Integral form

Let $n \in \mathbb{N}$, and let f be a function having $n+1$ *continuous derivatives on an open interval* I. Let $a, x \in I$. Then

$$f(x) = P_{n,a}(x) + R_{n,a}(x),$$

where

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)(x-t)^n}{n!}\, dt.$$

$R_{n,a}(x)$ is called the n-th remainder term.

Proof. Define

$$L_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \int_a^x \frac{f^{(n+1)}(t)(x-t)^n}{n!}\, dt.$$

We wish to prove $f(x) = L_n(x)$ for all $n \in \mathbb{N}$ and $x \in I$.

Proof by induction: Base case: $n = 0$. First note that if f' is continuous, the Fundamental Theorem of Calculus tells us that

$$f(x) = f(a) + \int_a^x f'(t)\, dt = L_0(x). \tag{1}$$

Induction Hypothesis: Assume $f(x) = L_m(x)$ for some m.

We wish to prove $f(x) = L_{m+1}(x)$.

$$L_{m+1}(x) = \sum_{k=0}^{m+1} \frac{f^{(k)}(a)}{k!}(x-a)^k + \int_a^x \frac{f^{(m+2)}(t)(x-t)^{m+1}}{(m+1)!}\, dt \tag{2}$$

Integrate by parts (possible because $f^{(m+2)}$ is cts by assumption).

Let

$$u(t) = \frac{(x-t)^{m+1}}{(m+1)!}, \quad u'(t) = -\frac{(x-t)^m}{m!},$$
$$v'(t) = f^{(m+2)}(t), \quad v(t) = f^{(m+1)}(t).$$

and then equation (2) becomes

$$\begin{aligned}
L_{m+1}(x) &= \sum_{k=0}^{m+1} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{(x-t)^{m+1}}{(m+1)!} f^{(m+1)}(t)\Big|_a^x - \int_a^x \frac{-(x-t)^m}{m!} f^{(m+1)}(t)\, dt \\
&= \sum_{k=0}^{m+1} \frac{f^{(k)}(a)}{k!}(x-a)^k - \frac{(x-a)^{m+1}}{(m+1)!} f^{(m+1)}(a) + \int_a^x \frac{(x-t)^m}{m!} f^{(m+1)}(t)\, dt \\
&= \sum_{k=0}^{m} \frac{f^{(k)}(a)}{k!}(x-a)^k + \int_a^x \frac{(x-t)^m}{m!} f^{(m+1)}(t)\, dt \\
&= L_m(x) = f(x) \quad \text{(by the induction hypothesis)},
\end{aligned}$$

□

Lemma 2.3.6

Suppose f is defined on an open interval I, and has continuous derivatives of all orders (infinitely differentiable) at $a \in I$.
Then the $P_{n,a}(x)$ converges to $f(x)$ iff $R_{n,a}(x)$ converges to 0.

Proof.

$\Rightarrow$: we write

$$R_{n,a}(x) = f(x) - P_{n,a}(x)$$

If

$$\lim_{n\to\infty} P_{n,a}(x) = f(x),$$

then

$$\begin{aligned}\lim_{n\to\infty} R_{n,a}(x) &= \lim_{n\to\infty} (f(x) - P_{n,a}(x)) \\ &= \lim_{n\to\infty} f(x) - \lim_{n\to\infty} P_{n,a}(x) \\ &= f(x) - f(x) = 0 \\ \Leftarrow: \quad P_{n,a}(x) &= f(x) - R_{n,a}(x)\end{aligned}$$

If

$$\lim_{n\to\infty} R_{n,a}(x) = 0,$$

then

$$\begin{aligned}\lim_{n\to\infty} P_{n,a}(x) &= \lim_{n\to\infty} (f(x) - R_{n,a}(x)) \\ &= \left(\lim_{n\to\infty} f(x)\right) - \left(\lim_{n\to\infty} R_{n,a}(x)\right) \\ &= f(x) - 0 = f(x).\end{aligned}$$

□

Definition 2.3.7

If $P_{n,a}(x)$ converges to $f(x)$ for all x in an open interval containing a, f is described as analytic at a.

Lemma 2.3.8: Lagrange form of remainder

If $a \neq x$, the n-th remainder term (from Taylor's theorem - Integral form) can be written as

$$R_{n,a}(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-a)^{n+1}$$

for some z between a and x.

Proof. Assume $x > a$ (The $x < a$ case can be done very similarly). Since $f^{(n+1)}$ is continuous on $[a, x]$, there exists (by the EVT):

$$m = \min\left\{f^{(n+1)}(t) \;\middle|\; t \in [a,x]\right\}$$

$$\mathcal{M} = \max\left\{f^{(n+1)}(t) \;\middle|\; t \in [a,x]\right\}$$

Since

$$0 \leq \frac{(x-t)^n}{n!} \quad \text{for } t \in [a,x],$$

we have

$$m\frac{(x-t)^n}{n!} \leq f^{(n+1)}(t)\frac{(x-t)^n}{n!} \leq \mathcal{M}\frac{(x-t)^n}{n!}$$

thus

$$m\int_a^x \frac{(x-t)^n}{n!}\,dt \leq R_{n,a}(x) \leq \mathcal{M}\int_a^x \frac{(x-t)^n}{n!}\,dt$$

$$\Rightarrow \quad m \leq R_{n,a}(x)\frac{(n+1)!}{(x-a)^{n+1}} \leq \mathcal{M}$$

By the I.V.T., there exists $z \in (a, x)$ such that

$$f^{(n+1)}(z) = \frac{(n+1)!}{(x-a)^{n+1}}R_{n,a}(x)$$

thus the result follows. □

Example.

> *The following are familiar Maclaurin series (you can establish these by considering $f^{(n)}(0)$ for each function).*
>
> $$\sin(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!} + \frac{\sin^{(2n+2)}(z)}{(2n+2)!} x^{2n+2} \quad \text{for some } z \text{ between } 0 \text{ and } x$$
>
> $$\cos(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{(2k)!} + \frac{\cos^{(2n+1)}(z)}{(2n+1)!} x^{2n+1} \quad \text{for some } z \text{ between } 0 \text{ and } x$$
>
> $$\exp(x) = \sum_{k=0}^{n} \frac{x^k}{k!} + \frac{\exp^{(n+1)}(z)}{(n+1)!} x^{n+1} \quad \text{for some } z \text{ between } 0 \text{ and } x.$$
>
> *Question: Are these functions analytic at 0?*

Proof. It's easy to estimate the first two.

$$\left|\sin^{(2n+2)}(z)\right| \leq 1 \quad \forall z$$

$$\left|\cos^{(2n+2)}(z)\right| \leq 1 \quad \forall z$$

Hence,

$$\left|\frac{\sin^{(2n+2)}(z)}{(2n+2)!} x^{2n+2}\right| \leq \frac{|x|^{2n+2}}{(2n+2)!}$$

$$\left|\frac{\cos^{(2n+2)}(z)}{(2n+1)!} x^{2n+1}\right| \leq \frac{|x|^{2n+1}}{(2n+1)!}$$

Both go to 0 as $n \to \infty$.

Hence, by the squeeze theorem,

$$\lim_{n\to\infty} \frac{\sin^{(2n+2)}(z) x^{2n+2}}{(2n+2)!} = \lim_{n\to\infty} \frac{\cos^{(2n+1)}(z) x^{2n+1}}{(2n+1)!} = 0$$

exp(x) is only slightly harder. We show when $x > 0$. Then z is in $(0, x)$ and

$$\exp^{(n+1)}(z) = \exp(z) < \exp(x) = e^x < 3^x$$

Hence,

$$\frac{\exp^{(n+1)}(z)\, x^{n+1}}{(n+1)!} < \frac{3^x\, x^{n+1}}{(n+1)!}$$
$$\Rightarrow \quad 0 \quad \text{as} \quad n \to \infty.$$

How large n must be will depend on x. □

Example.

> Can you show $R_{n,0}(x) \to 0$ when $x < 0$?

Proof. When $x < 0$, we have $x < z < 0$. Since exp is increasing, we have $\exp(z) < \exp(0) = 1$.

Thus,

$$|R_{n,0}(x)| < \frac{1}{(n+1)!} |x^{n+1}|$$

which goes to 0 as $n \to \infty$. □

Thus, we know e is analytic at all $a \in \mathbb{R}$.

Example.

Show that $e^{i\theta} = \cos\theta + i\sin\theta$.

Proof.

$$\text{Show that T.S. on the LHS} = \text{T.S. on the RHS.}$$

□

Example.

Use a Maclaurin Series for e^x to approximate e with an error less than 0.000005.

We know that

$$\begin{aligned}\exp(1) &= \sum_{k=0}^{n} \frac{1^k}{k!} + \frac{\exp^{(n+1)}(z)}{(n+1)!}1^{n+1} \quad \text{for } z \in [0,1] \\ &= 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{n!} + \frac{e^z}{(n+1)!}\end{aligned}$$

Proof. Since $0 < z < 1$,

$$1 = e^0 < e^z < e^1 = e < 3$$

This means the error term will be

$$\frac{e^z}{(n+1)!} < \frac{3}{(n+1)!}.$$

We need this to be < 0.000005.

Taking $n = 9$ suffices (check your calculator).

This lets us calculate:

$$e = 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!} + R_{9,0}(1)$$

where

$$0 < R_{9,0}(1) < 0.000005$$

so

$$e \in \left(1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!},\ 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!} + 0.000005\right)$$

□

Example.

e is irrational

Proof. We know that for any $n \in \mathbb{N}$,

$$e = 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{n!} + R_n,$$

$$0 < R_n < \frac{3}{(n+1)!}.$$

Suppose e were rational, so $e = \frac{a}{b}$, with a, b positive integers.

Choose $n > \max\{3, b\}$. So,

$$\frac{a}{b} = 1 + 1 + \frac{1}{2} + \cdots + \frac{1}{n!} + R_n$$

$$n!\frac{a}{b} = n! + n! + \frac{n!}{2} + \cdots + 1 + n!R_n$$

$n!R_n$ is an integer, since LHS is an integer.

$$0 < R_n < \frac{3}{(n+1)!}$$

thus

$$0 < n!R_n < \frac{3}{n+1} < \frac{3}{4}$$

no integer in this range. Contradiction! □

Example.

Let's compute the Taylor series for $f(x) = \ln(x)$ about $a = 1$.

Proof. *We need to know the derivatives at $x = 1$.*

$$\ln(1) = 0$$

$$\ln'(x)\Big|_{x=1} = x^{-1}\Big|_{x=1} = 1$$

$$\ln''(x)\Big|_{x=1} = -x^{-2}\Big|_{x=1} = -1$$

$$\ln'''(x)\Big|_{x=1} = 2x^{-3}\Big|_{x=1} = 2$$

$$\ln^{(4)}(x)\Big|_{x=1} = (2)(-3)x^{-4}\Big|_{x=1} = (2)(-3)$$

$$\vdots$$

Thus,

$$\ln^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{x^n}, \quad \ln^{(n)}(1) = (-1)^{n-1}(n-1)!.$$

And so,

$$\ln x = \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k}(x-1)^k + \frac{\ln^{(n+1)}(z)}{(n+1)!}(x-1)^{n+1}$$

for some z between 1 and x.

Let's take $n = 3$.

$$\ln x = 1(x-1) + \frac{(-1)}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \frac{(-1)}{4}\left(\frac{(x-1)}{z}\right)^4$$

So, for $x = 1.1$, we find

$$\begin{aligned}\ln(1.1) &= 0.1 - \frac{1}{2}(0.1)^2 + \frac{1}{3}(0.1)^3 - \frac{1}{4}\left(\frac{0.1}{z}\right)^4 \\ &= 0.09533\overline{3} + \left(\frac{-1}{4}\left(\frac{0.1}{z}\right)^4\right)\end{aligned}$$

Since $z > 1$, we know

$$\begin{aligned}|R_{3,1}(1.1)| &= \left|\frac{-1}{4z^4}0.1^4\right| < \left|\frac{-1}{4}0.1^4\right| \\ &= 0.0000025\end{aligned}$$

Since $R_{3,1}(1.1) < 0$, we have

$$\ln(1.1) \in \left(0.09533\overline{3} - 0.0000025,\ 0.09533\overline{3}\right).$$

□

Example.

Q: Is ln analytic at 1? If so, for what interval around 1 does the remainder converge to 0?

It is known that

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}(x-1)^k$$

converges for $(0, 2]$.

But does it converge to $\ln x$ in this interval?

Example.

Claim: ln is analytic at all $a \in (0, 2]$

Proof. Let $1 < x \leq 2$. Then since $1 < z < x$, we have

$$\left|\frac{(-1)^n}{n+1}\left(\frac{x-1}{z}\right)^{n+1}\right| \leq \frac{(x-1)^{n+1}}{n+1} \to 0 \quad \text{as } n \to \infty.$$

So if $1 < x \leq 2$, the Taylor series converges to $\ln x$.

Things are not so straightforward for $x < 1$. For this, we need the integral form of the remainder. We write

$$\begin{aligned}\ln(x) &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k}(x-1)^k + \int_1^x \frac{\ln^{(n+1)}(t)(x-t)^n}{n!}\,dt \\ &= \sum_{k=1}^{n} \frac{(-1)^{k-1}}{k}(x-1)^k + (-1)^n \int_1^x \frac{n!}{t^{n+1}} \cdot \frac{(x-t)^n}{n!}\,dt.\end{aligned}$$

So

$$R_{n,1}(x) = (-1)^n \int_1^x \frac{n!}{t^{n+1}} \cdot \frac{(x-t)^n}{n!}\,dt.$$

To investigate this integral, set $u = \frac{x-t}{t}$. If $t = 1$, then $u = x - 1$. If $t = x$, then $u = 0$. Also,

$$tu = x - t \Rightarrow (u+1)t = x \Rightarrow t = \frac{x}{u+1} \Rightarrow \frac{dt}{du} = -\frac{x}{(u+1)^2} \Rightarrow \frac{1}{t}\frac{dt}{du} = -\frac{1}{u+1}.$$

Hence, by u-substitution,

$$\begin{aligned}R_{n,1}(x) &= (-1)^n \int_{x-1}^{0} u^n \left(-\frac{1}{u+1}\right) du \\ &= (-1)^{n+1} \int_{x-1}^{0} \frac{u^n}{u+1}\,du.\end{aligned}$$

Since $0 < x < 1$, then since $x \leq t \leq 1$,

$$u = \frac{x-t}{t} \geq x - t \geq x - 1, \quad u = \frac{x-t}{t} \leq 0 \Rightarrow x - 1 \leq u \leq 0 \Rightarrow \frac{1}{x} \geq \frac{1}{u+1}.$$

Hence,

$$|R_{n,1}(x)| = \left|\int_{x-1}^{0} \frac{u^n}{u+1}\,du\right| \leq \int_{x-1}^{0} \left|\frac{u^n}{u+1}\right| du \leq \int_{x-1}^{0} \frac{|u|^n}{x}\,du = \frac{1}{|x|}\int_{x-1}^{0} |u|^n\,du.$$

Setting $v = -u$, $dv = -du$, $|u| = v$, so:

$$u = x-1 \Rightarrow v = 1-x, \quad u = 0 \Rightarrow v = 0 \Rightarrow |R_{n,1}(x)| \leq \frac{1}{|x|}\int_0^{1-x} v^n\,dv = \frac{1}{|x|}\cdot\frac{(1-x)^{n+1}}{n+1} \to 0 \quad \text{as } n \to \infty.$$

Hence, the Taylor series converges to $\ln x$ whenever the Taylor series converges.

□

Example.

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x > 0, \\ 0, & x \leq 0 \end{cases}$$

Proof. It can be shown that

$$f^{(n)}(x) = \begin{cases} g_n(x)e^{-\frac{1}{x^2}}, & x > 0, \\ 0, & x \leq 0 \end{cases} \quad \text{(check)}$$

where $g_n(x)$ is a polynomial in $1/x$ given recursively by

$$g_0(x) = 1$$

$$g_{n+1}(x) = g_n'(x) + \frac{2}{x^3}g_n(x).$$

Hence f is ∞ differentiable $\forall\, x$ with

$$f^{(n)}(0) = 0 \quad \forall\, n \geq 0$$

thus

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = 0$$

Thus, the Taylor series for f about 0 (Maclaurin series) exists, but it does <u>not</u> converge to f.

$$f(x) \neq \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \quad \text{when} \quad x > 0$$

□

Definition 2.3.9

A function of the form

$$f(x) = \sum_{n=0}^{\infty} a_n(x-b)^n = a_0 + a_1(x-b) + a_2(x-b)^2 + \cdots$$

is called a ***Power Series*** *centred at b.*

A power series may not converge for all values of x, so f may not be defined on $\mathbb{R}$. However, it will always be defined for $x = b$, since

$$f(b) = a_0 + a_1(b-b) + a_2(b-b)^2 + \cdots = a_0.$$

Theorem 2.3.10

Suppose that the series

$$\sum_{n=0}^{\infty} a_n x_0^n$$

converges, and let a be any number with $0 < a < |x_0|$. Then on $[-a, a]$ the series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges uniformly (and absolutely). Moreover, the same is true for the series

$$g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Finally, f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Proof. Since $\sum_{n=0}^{\infty} a_n x_0^n$ converges, $\lim_{n\to\infty} a_n x_0^n = 0$. Hence they are bounded: there is some number M such that

$$|a_n x_0^n| = |a_n||x_0|^n \leq M \quad \forall n.$$

Now, if x is in $[-a, a]$, then $|x| \leq |a|$, so

$$|a_n x^n| = |a_n||x^n| \leq |a_n||a|^n = |a_n||x_0|^n \left|\frac{a}{x_0}\right|^n \leq M \left|\frac{a}{x_0}\right|^n.$$

But $\left|\frac{a}{x_0}\right| < 1$, so the geometric series

$$\sum_{n=0}^{\infty} M \left|\frac{a}{x_0}\right|^n$$

converges.

Choosing $M_n = M \left|\frac{a}{x_0}\right|^n$, it follows from the Weierstrass M-test that $\sum_{n=0}^{\infty} a_n x^n$ converges uniformly on $[-a, a]$.

To prove the same assertion for g, notice that

$$\begin{aligned}
\left|n a_n x^{n-1}\right| &= n|a_n||x^{n-1}| \\
&\leq n|a_n||a|^{n-1} \\
&= |a_n||x_0|^n \frac{n}{|a|} \left|\frac{a}{x_0}\right|^n \\
&\leq \frac{M}{|a|} n \left|\frac{a}{x_0}\right|^n
\end{aligned}$$

Since $\left|\frac{a}{x_0}\right| < 1$, the series

$$\sum_{n=1}^{\infty} \frac{M}{|a|} n \left|\frac{a}{x_0}\right|^n$$

converges by the ratio test.

Another appeal to the W-M-test proves that

$$\sum_{n=1}^{\infty} n a_n x^{n-1}$$

converges uniformly on $[-a, a]$.

Finally, since

1. $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges uniformly,
2. $g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ converges uniformly,
3. $n a_n x^{n-1}$ is continuous for each n,

Theorem from the "sequences of functions" chapter tells us that

$$f'(x) = g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{for} \quad x \in [-a, a]$$

and

$$\frac{d}{dx} \sum_{n=0}^{\infty} a_n x^n = \sum_{n=1}^{\infty} \frac{d}{dx} \left(a_n x^n \right).$$

□

Theorem 2.3.11: Any power series which converges on an open interval is its own Taylor Series

If

$$f(x) = \sum_{n=0}^{\infty} a_n (x-b)^n$$

converges for all x in some interval $(-R+b, b+R)$, then f has derivatives of all orders in $(-R+b, b+R)$. Moreover,

$$a_k = \frac{f^{(k)}(b)}{k!}.$$

Proof. We first prove this when $b = 0$.

Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges on $(-R, R)$.

Applying theorem above,

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

also converges on $(-R, R)$.

Since f' is also a power series, applying theorem above again yields

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Applying it k times gives

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1) a_n x^{n-k}$$

$$= k! a_k + \sum_{n=k+1}^{\infty} n(n-1)\cdots(n-k+1) a_n x^{n-k}$$

on the same interval.

Thus f is differentiable at all orders on $(-R, R)$.

Moreover,

$$f^{(k)}(0) = k! a_k,$$

and the result follows.

If $b \neq 0$, define

$$\overline{f}(x) = \sum_{n=0}^{\infty} a_n x^n, \quad \text{and} \quad f(x) = \overline{f}(x-b).$$

It then follows that

$$f^{(k)}(b) = \overline{f}^{(k)}(b-b) = k! a_k.$$

$\square$

Chapter 3

Vector Functions

3.1 Preliminaries

Notation: If

$$\mathbf{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n, \quad \mathbf{y} = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$$

$$\implies (\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} x_i y_i$$

$$|\mathbf{x}| \equiv \|\mathbf{x}\| = (\mathbf{x}, \mathbf{x})^{1/2} = \left(\sum_{i=1}^{n} x_i x_i \right)^{1/2} \qquad \text{(the } \textbf{norm} \text{ of } \mathbf{x}\text{)}.$$

Note:

1. $|\mathbf{x}| = 0 \iff \mathbf{x} = 0$
2. $(\mathbf{x}, \mathbf{0}) = 0 \quad$ for all $\mathbf{x}$
3. In general $(\mathbf{x}, \mathbf{y}) = 0$ does **not** imply that $\mathbf{x} = 0$ or $\mathbf{y} = 0$.
   See for example, $\mathbf{x} = (1, 1)$, $\mathbf{y} = (1, -1)$.
4. If $s \in \mathbb{R}$, $\|s\underline{x}\| = |s|\|\underline{x}\|$.
5. $\forall \underline{x} \in \mathbb{R}^n$, $\|\underline{x}\| \geq 0$

Theorem 3.1.1: Cauchy-Schwarz Inequality

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, *then*

$$|(\mathbf{x}, \mathbf{y})| \leq |\mathbf{x}||\mathbf{y}|$$

Proof. Trivially true when $\mathbf{x} = 0$ or $\mathbf{y} = 0$. So assume $\mathbf{x}, \mathbf{y} \neq 0$. Let $\lambda \in \mathbb{R}$ and

$$f(\lambda) = \sum_{i=1}^{n} (\lambda x_i + y_i)^2 \geq 0 \quad \text{(sum of squares of real numbers)}$$

Then

$$\begin{aligned} f(\lambda) &= \sum_{i=1}^{n} \lambda^2 x_i^2 + 2\lambda x_i y_i + y_i^2 \\ &= \lambda^2 \left(\sum_{i=1}^{n} x_i^2 \right) + 2\lambda \sum_{i=1}^{n} (x_i y_i) + \sum_{i=1}^{n} y_i^2. \end{aligned}$$

Now, if

$$a = \sum_{i=1}^{n} x_i^2, \quad b = 2\sum_{i=1}^{n} x_i y_i, \quad c = \sum_{i=1}^{n} y_i^2,$$

Then $a, c > 0$ and we have the quadratic:

$$f(\lambda) = a\lambda^2 + b\lambda + c \geq 0$$

If $b^2 - 4ac > 0$, f has two distinct real roots. But this implies f is strictly negative between λ_1, λ_2 (contradiction). Hence,

$$b^2 - 4ac \leq 0$$

Thus,

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 - 4\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) \leq 0$$

$$(\mathbf{x}, \mathbf{y})^2 \leq |\mathbf{x}|^2 |\mathbf{y}|^2 \quad \text{Result follows.}$$

□

Theorem 3.1.2: Triangle Inequality

If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, *then*

$$|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$$

Proof.

$$\begin{aligned}
|\mathbf{x} + \mathbf{y}|^2 &= (\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}) \\
&= (\mathbf{x}, \mathbf{x}) + (\mathbf{y}, \mathbf{y}) + (\mathbf{x}, \mathbf{y}) + (\mathbf{y}, \mathbf{x}) \\
&= |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2(\mathbf{x}, \mathbf{y}) \\
&\leq |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2|(\mathbf{x}, \mathbf{y})| \\
&\leq |\mathbf{x}|^2 + |\mathbf{y}|^2 + 2|\mathbf{x}||\mathbf{y}| \quad \text{(Cauchy-Schwarz)} \\
&= (|\mathbf{x}| + |\mathbf{y}|)^2
\end{aligned}$$

Take the square root of both sides, the result follows. □

3.2 Limits and continuity of functions

$$f : \mathbb{R}^n \to \mathbb{R}^m \quad \text{or} \quad f : \Omega \to \mathbb{R}^m \text{ where } \Omega \subseteq \mathbb{R}^n.$$

Every $f : \Omega \to \mathbb{R}^m$ *can be written as*

$$f = (f_1, f_2, \ldots, f_m)^T, \quad \text{where } f_i : \Omega \to \mathbb{R}, \text{ for all } i = 1, \ldots, m.$$

f_i *is a component of* f.

We can extract f_i *by taking the inner prodcut:*

$$f_i(\underline{x} = (f(\underline{x}), \underline{e_i})$$

Example.

$$f(x_1, x_2) = \left(\sqrt{1 - (x_1^2 + x_2^2)},\ x_1 + x_2\right)^T.$$

Here, $n = m = 2$. The domain of f is

$$\Omega = \left\{(x_1, x_2) \in \mathbb{R}^2 \;\middle|\; x_1^2 + x_2^2 \leq 1\right\}.$$

$$f : \Omega \to \mathbb{R}^2.$$

Definition 3.2.1

Suppose Ω is a subset of $\mathbb{R}^n$. The point $\mathbf{a} \in \mathbb{R}^n$ is a ***limit point*** *of Ω if for every $\varepsilon > 0$, there exists $\mathbf{y} \in \Omega$ such that $0 < |\mathbf{y} - \mathbf{a}| < \varepsilon$.*

Definition 3.2.2

Let $\mathbf{a} \in \mathbb{R}^n$.

1. The *open r ball* of $\mathbf{a}$ is defined as

$$B_r(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{a}| < r\}$$

2. The *closed r ball* of $\mathbf{a}$ is defined as

$$\overline{B}_r(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{a}| \leq r\}$$

Example.

$$\Omega = B_1(\underline{0}) = \{\mathbf{x} \in \mathbb{R}^n \text{ such that } |\mathbf{x}| < 1\}.$$

The set of limit points of Ω is $\{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x}| \leq 1\}$.

Example.

$$\Omega = B_1(\underline{0}) \setminus \{(0,0)\} \quad \text{in } \mathbb{R}^2.$$

The set of limit points of Ω is $\{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x}| \leq 1\}$.

Definition 3.2.3

Let $f : \Omega \to \mathbb{R}^m$ be a function (here $\Omega \subseteq \mathbb{R}^n$). Let $\mathbf{x}_0$ be a limit point of Ω. Let $\mathbf{y}_0 \in \mathbb{R}^m$. Then

$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = \mathbf{y}_0$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that if

$$\mathbf{x} \in \Omega \text{ and } 0 < |\mathbf{x} - \mathbf{x}_0| < \delta \quad \Rightarrow \quad |f(\mathbf{x}) - \mathbf{y}_0| < \epsilon.$$

Definition 3.2.4

Consider f defined as before. Then f is ***continuous*** *at $\mathbf{x}_0$ if*

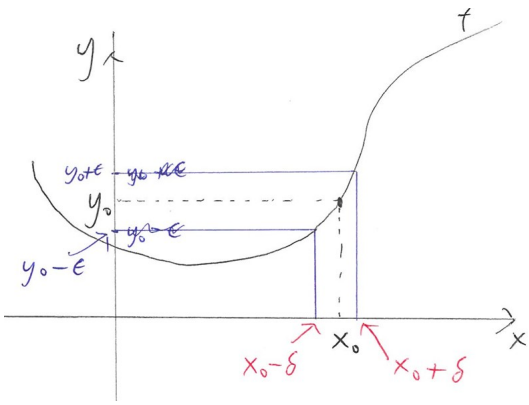
$$\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

This is equivalent to saying that for every $\epsilon > 0$, there exists $\delta > 0$ such that if

$$\mathbf{x} \in \Omega \text{ and } |\mathbf{x} - \mathbf{x}_0| < \delta \quad \Rightarrow \quad |f(\mathbf{x}) - f(\mathbf{x}_0)| < \epsilon.$$

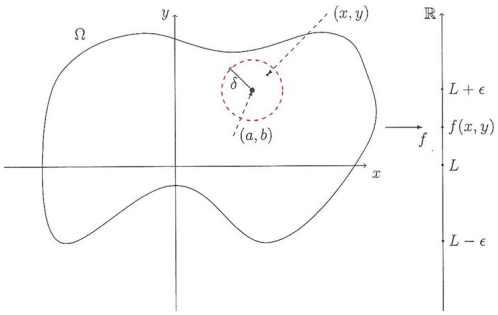
Example.

Suppose $n = 1$, $m = 1$. Then



Example.

Suppose $n = 2$, $m = 1$. Then



Example.

Suppose $n = 2$, $m = 2$. Then

Theorem 3.2.5: If the limit exists, it is unique

Suppose $\Omega \subseteq \mathbb{R}^n$, $\mathbf{a}$ is a limit point of Ω, and let $f : \Omega \to \mathbb{R}^m$. If

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = \mathbf{L}_1 \quad \textit{and} \quad \lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = \mathbf{L}_2, \quad \textit{then } \mathbf{L}_1 = \mathbf{L}_2.$$

Proof. The proof is very similar to the single variable case in the first half of the course. (not assessed) Fix $\varepsilon > 0$. Choose $\delta_1, \delta_2 > 0$ such that

$$0 < |\mathbf{x} - \mathbf{a}| < \delta_1 \quad \Rightarrow \quad |f(\mathbf{x}) - \mathbf{L}_1| < \frac{\varepsilon}{2},$$

$$0 < |\mathbf{x} - \mathbf{a}| < \delta_2 \quad \Rightarrow \quad |f(\mathbf{x}) - \mathbf{L}_2| < \frac{\varepsilon}{2}.$$

Choose $\delta = \min\{\delta_1, \delta_2\}$ and if $|\mathbf{x} - \mathbf{a}| < \delta$, then

$$\begin{aligned} |\mathbf{L}_1 - \mathbf{L}_2| &= |\mathbf{L}_1 - f(\mathbf{x}) + f(\mathbf{x}) - \mathbf{L}_2| \\ &\leq |\mathbf{L}_1 - f(\mathbf{x})| + |f(\mathbf{x}) - \mathbf{L}_2| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Since this holds for arbitrary $\varepsilon > 0$, we must have $|\mathbf{L}_1 - \mathbf{L}_2| = 0$, so $\mathbf{L}_1 = \mathbf{L}_2$.

$\square$

Theorem 3.2.6

Suppose $\Omega \subseteq \mathbb{R}^n$, $c \in \mathbb{R}$, $\mathbf{a}$ is a limit point of Ω, and let $f, g : \Omega \to \mathbb{R}^m$. If $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})$, $\lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x})$ both exist, then

1. $\lim\limits_{\mathbf{x}\to\mathbf{a}} (f+g)(\mathbf{x}) = \left(\lim\limits_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})\right) + \left(\lim\limits_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x})\right)$
2. $\lim\limits_{\mathbf{x}\to\mathbf{a}} (cf)(\mathbf{x}) = c \lim\limits_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x})$

Proof. The proof is very similar to the single variable case in the first half of the course. (not assessed) Suppose

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = \mathbf{b}_1 \quad \text{and} \quad \lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x}) = \mathbf{b}_2.$$

1. Fix $\varepsilon > 0$. Choose $\delta_1, \delta_2 > 0$ such that

$$0 < |\mathbf{x} - \mathbf{a}| < \delta_1 \Rightarrow |f(\mathbf{x}) - \mathbf{b}_1| < \frac{\varepsilon}{2},$$

$$0 < |\mathbf{x} - \mathbf{a}| < \delta_2 \Rightarrow |g(\mathbf{x}) - \mathbf{b}_2| < \frac{\varepsilon}{2}.$$

   Choose $\delta = \min\{\delta_1, \delta_2\}$ and if $|\mathbf{x} - \mathbf{a}| < \delta$, then

$$\begin{aligned} |(f+g)(\mathbf{x}) - (\mathbf{b}_1 + \mathbf{b}_2)| &= |f(\mathbf{x}) - \mathbf{b}_1 + g(\mathbf{x}) - \mathbf{b}_2| \\ &\leq |f(\mathbf{x}) - \mathbf{b}_1| + |g(\mathbf{x}) - \mathbf{b}_2| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

2. The claim is trivially true when $c = 0$. Suppose $c \neq 0$. Fix $\varepsilon > 0$. There exists $\delta > 0$ such that

$$0 < |\mathbf{x} - \mathbf{a}| < \delta \Rightarrow |f(\mathbf{x}) - \mathbf{b}_1| < \frac{\varepsilon}{|c|}.$$

   Then

$$|cf(\mathbf{x}) - c\mathbf{b}_1| = |c|\,|f(\mathbf{x}) - \mathbf{b}_1| < |c| \cdot \frac{\varepsilon}{|c|} = \varepsilon.$$

□

Theorem 3.2.7: Squeeze theorem

Suppose $\Omega \subseteq \mathbb{R}^n$, $\mathbf{a}$ is a limit point of Ω, and let $f, g, h : \Omega \to \mathbb{R}$. Let $f(\mathbf{x}) \leq g(\mathbf{x}) \leq h(\mathbf{x})$ for all $\mathbf{x}$ on some $\delta > 0$ ball of $\mathbf{a}$. If the limits of f and h exist with

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = \lim_{\mathbf{x}\to\mathbf{a}} h(\mathbf{x}) = L,$$

then the limit of g likewise exists and

$$\lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x}) = L.$$

Proof. The proof is very similar to the single variable case in the first half of the course. (not assessed) We suppose

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = \lim_{\mathbf{x}\to\mathbf{a}} h(\mathbf{x}) = L,$$

and

$$f(\mathbf{x}) \leq g(\mathbf{x}) \leq h(\mathbf{x}) \quad \text{for all } \mathbf{x} \in B_{\delta_1}(\mathbf{a}) \text{ for some } \delta_1 > 0.$$

Let $\varepsilon > 0$, and choose $\delta_2 > 0$ such that for all $\mathbf{x} \in B_{\delta_1}(\mathbf{a}) \setminus \{\mathbf{a}\}$,

$$|f(\mathbf{x}) - L| < \varepsilon \tag{8}$$

$$|h(\mathbf{x}) - L| < \varepsilon. \tag{9}$$

Choose $\delta = \min\{\delta_1, \delta_2\}$, and note that if $\mathbf{x} \in B_\delta(\mathbf{a}) \setminus \{\mathbf{a}\}$,

$$f(\mathbf{x}) \leq g(\mathbf{x}) \leq h(\mathbf{x}) \quad \Rightarrow \quad f(\mathbf{x}) - L \leq g(\mathbf{x}) - L \leq h(\mathbf{x}) - L.$$

Using equations (8) and (9), we have

$$-\varepsilon < f(\mathbf{x}) - L \leq g(\mathbf{x}) - L \leq h(\mathbf{x}) - L < \varepsilon \quad \Rightarrow \quad |g(\mathbf{x}) - L| < \varepsilon.$$

Hence, $\lim_{\mathbf{x}\to\mathbf{a}} g(\mathbf{x}) = L$. □

Lemma 3.2.8

The function $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ satisfies

$$\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b} = (b_1, b_2, \ldots, b_m)$$

if and only if $\lim_{\mathbf{x}\to\mathbf{x}_0} f_i(\mathbf{x}) = b_i$ *for* $i = 1, \ldots, m$.

Proof. $\Rightarrow$ Suppose $\lim_{x\to x_0} f(x) = b$.
Therefore, $\forall\, \varepsilon > 0$, $\exists\, \delta > 0$ such that if $x \in U$ and $0 < |x - x_0| < \delta \;\Rightarrow\; \|f(x) - b\| < \varepsilon$.
Let $i \in \{1, \ldots, m\}$ and let ε, δ be as above. Take $x \in U$ and $0 < |x - x_0| < \delta$. Then

$$|f_i(x) - b_i| = \sqrt{(f_i(x) - b_i)^2} \leq \sqrt{\sum_{j=1}^{m} (f_j(x) - b_j)^2} = \|f(x) - b\| < \varepsilon.$$

$\Leftarrow$ Suppose $\lim_{x\to x_0} f_i(x) = b_i, \quad \forall\, i \in \{1, \ldots, m\}$. Then fix $\varepsilon > 0$ and $\forall\, i$, $\exists\, \delta_i > 0$ such that if $x \in U$ and $0 < |x - x_0| < \delta_i$,

$$|f_i(x) - b_i| < \frac{\varepsilon}{\sqrt{m}}.$$

Take $\delta = \min\{\delta_1, \delta_2, \ldots, \delta_m\}$. Then if $x \in U$ and $0 < |x - x_0| < \delta$,

$$\|f(x) - b\|^2 = \sum_{i=1}^{m} |f_i(x) - b_i|^2 < m \left(\frac{\varepsilon}{\sqrt{m}}\right)^2 = \varepsilon^2.$$

Take the square root of both sides, result follows. □

Theorem 3.2.9

The function $f : \Omega \to \mathbb{R}^m$ is continuous at $\mathbf{x}_0$ if and only if $f_1, f_2, \ldots, f_m$ are continuous at $\mathbf{x}_0$.

Example.

Consider $f : \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$ such that

$$f(x, y) = \frac{xy}{\sqrt{x^2 + y^2}}.$$

Does

$$\lim_{(x,y)\to(0,0)} f(x, y)$$

exist?

Proof. *I claim it does. Let's prove it.*
Fix $\varepsilon > 0$. We want to find $\delta > 0$ such that if $0 < |(x, y) - (0, 0)| < \delta$, then

$$|f(x, y) - L| < \varepsilon$$

for some L.

Consider (x, y) such that

$$0 < |(x, y) - (0, 0)| = \sqrt{x^2 + y^2} < \delta.$$

Observe that

$$(|x| - |y|)^2 \geq 0 \Rightarrow x^2 + y^2 - 2|xy| \geq 0 \Rightarrow |xy| \leq \frac{x^2 + y^2}{2}.$$

Now,

$$\left| \frac{xy}{\sqrt{x^2 + y^2}} \right| = \frac{|xy|}{\sqrt{x^2 + y^2}} \leq \frac{x^2 + y^2}{2\sqrt{x^2 + y^2}} = \frac{1}{2}\sqrt{x^2 + y^2} < \frac{\delta}{2}.$$

If we take $\delta = 2\varepsilon$, then

$$\left| \frac{xy}{\sqrt{x^2 + y^2}} - 0 \right| < \varepsilon.$$

Thus,

$$\lim_{(x,y)\to(0,0)} f(x, y) = 0.$$

□

Example.

> *Consider* $f : \mathbb{R}^2 \setminus \{(0, 0)\} \to \mathbb{R}$ *such that*
>
> $$f(x, y) = \frac{xy^2}{x^2 + y^2}.$$
>
> ***Claim:***
>
> $$\lim_{(x,y)\to(0,0)} f(x, y) = 0.$$

Proof. Note that

$$|f(x, y) - 0| = \left| \frac{xy^2}{x^2 + y^2} \right| = |x| \cdot \frac{y^2}{x^2 + y^2} \leq |x| \quad (*)$$

If $0 < |(x, y) - (0, 0)| < \delta$, then

$$|x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2} < \delta \quad (**)$$

From (*) and (**), we get:

$$|f(x, y) - 0| \leq |x| < \delta.$$

Given any $\varepsilon > 0$, take $\delta = \varepsilon$. Then $|f(x, y) - 0| < \varepsilon$. The result follows. □

Example.

> Does $\lim_{(x,y)\to(0,0)} \frac{x^2+xy}{x^2+y^2}$ exist?

Proof. *Assume the limit exists and equals* L. Take $\varepsilon = \frac{1}{4}$. Then there exists $\delta > 0$ such that if $0 < |(x, y) - (0, 0)| < \delta$, then

$$\left| \frac{x^2 + xy}{x^2 + y^2} - L \right| < \frac{1}{4}.$$

Take $x = 0$, $y \neq 0$: then $0 < |(0, y)| < \delta$.

$$\left| \frac{x^2 + xy}{x^2 + y^2} - L \right| = \left| \frac{0}{y^2} - L \right| = |0 - L| = |L| < \frac{1}{4}.$$

Now take $x = y$: then $0 < |(x, x)| < \delta$.

$$\left|\frac{x^2 + xy}{x^2 + y^2} - L\right| = \left|\frac{x^2 + x^2}{x^2 + x^2} - L\right| = \left|\frac{2x^2}{2x^2} - L\right| = |1 - L| < \frac{1}{4}.$$

But this means:

$$-\frac{1}{4} < L < \frac{1}{4} \quad \text{and} \quad \frac{3}{4} < L < \frac{5}{4},$$

which is a contradiction. Thus, the limit does not exist.

□

Intuitively, we approach along the y-axis first, then $x = y$.

Theorem 3.2.10

Let $\Omega \subseteq \mathbb{R}^n$. Consider a multivariable function $f : \Omega \to \mathbb{R}^m$ which is continuous at $\mathbf{a} \in \Omega$. Let $g : U \subseteq \mathbb{R}^p \to \Omega$ satisfy

$$\lim_{\mathbf{t} \to \mathbf{k}} g(\mathbf{t}) = \mathbf{a}. \tag{10}$$

Then

$$\lim_{\mathbf{t} \to \mathbf{k}} (f \circ g)(\mathbf{t}) = f\left(\lim_{\mathbf{t} \to \mathbf{k}} g(\mathbf{t})\right) = f(\mathbf{a}).$$

Proof. We want to show that $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$$0 < |\mathbf{t} - \mathbf{k}| < \delta \quad \Rightarrow \quad |f(g(\mathbf{t})) - f(\mathbf{a})| < \varepsilon.$$

Fix $\varepsilon > 0$. Since f is continuous at $\mathbf{a}$, we have

$$\lim_{\mathbf{h} \to \mathbf{a}} f(\mathbf{h}) = f(\mathbf{a}).$$

This means there exists $\triangle > 0$ such that

$$|\mathbf{h} - \mathbf{a}| < \triangle \quad \Rightarrow \quad |f(\mathbf{h}) - f(\mathbf{a})| < \varepsilon. \tag{*}$$

Since $\triangle > 0$, equation (10) implies there exists $\delta > 0$ such that

$$0 < |\mathbf{t} - \mathbf{k}| < \delta \quad \Rightarrow \quad |g(\mathbf{t}) - \mathbf{a}| < \triangle. \tag{\#}$$

Taking (*) and (#) together, we get:

$$0 < |\mathbf{t} - \mathbf{k}| < \delta \quad \Rightarrow \quad |g(\mathbf{t}) - \mathbf{a}| < \triangle \quad \Rightarrow \quad |f(g(\mathbf{t})) - f(\mathbf{a})| < \varepsilon,$$

as required. □

Corollary 3.2.11

The two path rule: *Let $\Omega \subseteq \mathbb{R}^n$.* Consider a multivariable function $T : \Omega \to \mathbb{R}^m$. Suppose $\mu, \eta : U \subseteq \mathbb{R} \to \Omega$, with

$$\lim_{t \to k} \mu(t) = \lim_{t \to k} \eta(t) = \mathbf{a},$$

(but $\mu(t) \neq \mathbf{a}$, $\eta(t) \neq \mathbf{a}$ for $t \neq k$).
If

$$\lim_{t \to k} T(\mu(t)) \neq \lim_{t \to k} T(\eta(t)),$$

then T does not have a limit at $\mathbf{a}$.

Proof. Suppose $\lim_{\mathbf{h} \to \mathbf{a}} T = \mathbf{L}$. Then $\bar{T}$ defined by

$$\bar{T}(\mathbf{h}) = \begin{cases} T(\mathbf{h}), & \text{for } \mathbf{h} \neq \mathbf{a}, \\ \mathbf{L}, & \text{for } \mathbf{h} = \mathbf{a} \end{cases}$$

is continuous. If $t \neq k$, then $\eta(t) \neq \mathbf{a}$, and so $\bar{T}(\eta(t)) = T(\eta(t))$. Hence,

$$\lim_{t\to k} T(\eta(t)) = \lim_{t\to k} \bar{T}(\eta(t)) \quad (\text{since } t \neq k) = \bar{T}(\mathbf{a}) \quad (\text{by continuity}) = \mathbf{L}.$$

Similarly,

$$\lim_{t\to k} T(\mu(t)) = \mathbf{L}.$$

Hence, if

$$\lim_{t\to k} T(\eta(t)) \neq \lim_{t\to k} T(\mu(t)),$$

Contradiction. No limit exists. □

Example.

> *Consider*
>
> $$f(x,y,z) = \frac{xy + y^2 + zx}{x^2 + y^2 + z^2}.$$
>
> *Does the following limit exist?*
>
> $$\lim_{(x,y,z)\to(0,0,0)} f(x,y,z).$$

Proof. *Along the z-axis:* $x = y = 0$. Let $x(t) = 0$, $y(t) = 0$, $z(t) = t$, then

$$\lim_{t\to 0} f(0,0,t) = \lim_{t\to 0} \frac{0+0+0}{0+0+t^2} = \lim_{t\to 0} 0 = 0.$$

Along $x = y = z$*:* Let $x(t) = y(t) = z(t) = t$, then

$$\lim_{t\to 0} f(t,t,t) = \lim_{t\to 0} \frac{t\cdot t + t^2 + t\cdot t}{t^2 + t^2 + t^2} = \lim_{t\to 0} \frac{3t^2}{3t^2} = 1.$$

Thus, the limit does not exist. □

Example.

> *The function* $f : \mathbb{R}^2 \setminus \{(x,y) | y = x\} \to \mathbb{R}$ *is defined by*
>
> $$f(x,y) = \frac{x^2}{y-x}.$$
>
> *It can be shown that for all* $m \in \mathbb{R}$ *and all* $\alpha > 0$,
>
> $$\lim_{t\to 0} f(t, mt^\alpha) = 0,$$
>
> *(so long as* f *is defined). Does* $\lim_{(x,y)\to(0,0)} f(x,y)$ *exist?*

Proof. **a)** Clearly $f(x, 1x^1)$ is not defined. For values of m and α which are not both 1, we have

$$f(x, mx^\alpha) = \frac{x^2}{mx^\alpha - x}.$$

To calculate this limit, observe that

$$\lim_{x\to 0} x^2 = \lim_{x\to 0} mx^\alpha - x = 0.$$

Thus, we use L'Hôpital's Rule:

$$\lim_{x\to 0} \frac{x^2}{mx^\alpha - x} = \lim_{x\to 0} \frac{2x}{m\alpha x^{\alpha-1} - 1}. \tag{7}$$

If $\alpha \geq 1$, (7) becomes

$$\lim_{x\to 0} \frac{2x}{m\alpha x^{\alpha-1} - 1} = \frac{0}{-1} = 0.$$

If $0 < \alpha < 1$, we multiply (7) top and bottom by $x^{1-\alpha}$:

$$\lim_{x\to 0} \frac{2x}{m\alpha x^{\alpha-1} - 1} = \lim_{x\to 0} \frac{2x^{2-\alpha}}{m\alpha - x^{1-\alpha}} = \frac{0}{m\alpha} = 0.$$

Thus, the limit is always 0 along these paths.

b) To find contours of f, we fix a constant z_0 and write

$$z_0 = f(x, y).$$

If $z_0 = 0$, we have:

$$0 = \frac{x^2}{y - x} \quad \Rightarrow \quad x = 0.$$

If $z_0 \neq 0$, then

$$z_0 = \frac{x^2}{y - x} \Rightarrow z_0 y - z_0 x = x^2 \Rightarrow y = x + \frac{x^2}{z_0}.$$

These contours are all parabolas which pass through the origin.

c) In part (a), we established that for a wide variety of paths, the limit was 0. Using (b), we recall that contours $z_0 \neq 0$ correspond to parabolas of the form $y = x + \frac{x^2}{z_0}$.

But these parabolas go through the origin, which is the point whose limit we are seeking to establish or rule out!

Let's travel along such a contour. Note that $y - x = \frac{x^2}{z_0} \neq 0$ except at the origin. Thus, we are *never* passing through the line $y = x$, and:

$$f(t, t + \tfrac{t^2}{z_0}) = \frac{t^2}{(t + \frac{t^2}{z_0}) - t} = \frac{t^2}{\frac{t^2}{z_0}} = z_0.$$

This is not surprising, since we established in part (b) that such a curve was a contour with height z_0.

But then:

$$\lim_{t\to 0} f\left(t, t + \frac{t^2}{z_0}\right) = \lim_{t\to 0} z_0 = z_0.$$

To be specific, we can choose $z_0 = 1$, and write:

$$\lim_{t\to 0} f\left(t, t + \frac{t^2}{1}\right) = \lim_{t\to 0} 1 \neq 0.$$

We have thus found our counterexample path. The limit does not exist!

□

Chapter 4

Multivariable Differentiation

4.1 Preliminaries

Lemma 4.1.1

If $T \in M_{n\times m}$, there is $M \geq 0$ such that $|T\mathbf{h}| \leq M|\mathbf{h}|$ for $\mathbf{h} \in \mathbb{R}^n$.

Proof. If $\mathbf{h} = 0$, then $|T\mathbf{h}| = 0 = M|\mathbf{h}|$. Suppose $\mathbf{h} \neq 0$. Then $|\mathbf{h}| \neq 0$ and

$$\left|T\left(\frac{\mathbf{h}}{|\mathbf{h}|}\right)\right| = \left|T\left(\sum_{i=1}^{n}\frac{h_i}{|\mathbf{h}|}\mathbf{e}_i\right)\right| = \left|\sum_{i=1}^{n}\frac{h_i}{|\mathbf{h}|}T(\mathbf{e}_i)\right| \leq \sum_{i=1}^{n}\left|\frac{h_i}{|\mathbf{h}|}\right||T(\mathbf{e}_i)| \quad \text{(triangle inequality)}$$

$$= \sum_{i=1}^{n}\frac{|h_i|}{|\mathbf{h}|}|T(\mathbf{e}_i)| \leq \sum_{i=1}^{n}|T(\mathbf{e}_i)|.$$

Let $k = \max\{|T(\mathbf{e}_i)|\}$, then

$$\left|T\left(\frac{\mathbf{h}}{|\mathbf{h}|}\right)\right| \leq \sum_{i=1}^{n} k = nk.$$

Set $M = nk$. Then

$$\left|T\left(\frac{\mathbf{h}}{|\mathbf{h}|}\right)\right| = \frac{1}{|\mathbf{h}|}|T(\mathbf{h})| \leq M \Rightarrow |T(\mathbf{h})| \leq M|\mathbf{h}|.$$

□

4.2 Differentiability

Recall that in single variable calculus, a function f is differentiable at a if there exists a number $f'(a)$ such that

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}.$$

This can be re-written as

$$\lim_{h\to 0}\frac{f(a+h)-f(a)-f'(a)h}{h} = 0.$$

Informally, we say that f is differentiable at a if there is a straight line

$$y(a+h) = f(a) + f'(a)h$$

such that the difference $f(a+h) - y(a+h)$ goes to 0 faster than linearly in the limit as h goes to 0.

Definition 4.2.1

Suppose $\Omega \subseteq \mathbb{R}^n$. We say Ω is open if for every $\mathbf{x} \in \Omega$, there exists $\delta > 0$ such that if $|\mathbf{x} - \mathbf{y}| < \delta$, then $\mathbf{y} \in \Omega$.

Extending to the multivariable case, let $\Omega \subseteq \mathbb{R}^n$ be open.

Definition 4.2.2

*Let $a \in \Omega$. A function $f : \Omega \to \mathbb{R}^m$ is **differentiable** at a if there exists a matrix $J \in M_{m\times n}$ such that*

$$\lim_{h\to 0} \frac{|f(a+h) - f(a) - Jh|}{|h|} = 0.$$

*In this case, J is the **derivative** of f at a (we can also write $J = f'(a) \equiv Df(a)$).*
Informally, we say that f is differentiable at a if there is a linear transformation

$$y(a+h) = f(a) + f'(a)h$$

such that the difference $f(a+h) - y(a+h)$ goes to $\mathbf{0}$ faster than linearly in the limit as $\mathbf{h}$ goes to $\mathbf{0}$.

It is understood that $\mathbf{h} \in \mathbb{R}^n$. If $|\mathbf{h}|$ is small enough, then $\mathbf{a} + \mathbf{h} \in \Omega$, since Ω is open.

Definition 4.2.3

*When f is differentiable at every $\mathbf{x} \in \Omega$, we say simply that f is **differentiable**.*

Example.

> *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ such that*
>
> $$f(\mathbf{x}) = B\mathbf{x},$$
>
> *where $B \in M_{m\times n}$. Then f is differentiable, and $f'(a) = B$.*

Proof. To see this, write

$$\frac{|f(a+h) - f(a) - Bh|}{|h|} = \frac{|B(a+h) - Ba - Bh|}{|h|} = \frac{|Ba + Bh - Ba - Bh|}{|h|} = \frac{|0|}{|h|} = 0.$$

Hence,

$$\lim_{h\to 0} \frac{|f(a+h) - f(a) - Bh|}{|h|} = 0.$$

If $n = m = 1$, $f(x) = bx$, $b \in \mathbb{R}$, then for $a \in \mathbb{R}$, we have $f'(a) = b$. □

Theorem 4.2.4: Differentiable implies continuous

*Let $\Omega \subseteq \mathbb{R}^n$ be open, and suppose function $f : \Omega \to \mathbb{R}^m$ is **differentiable** at a. Then f is continuous at a.*

Proof. We want to show that

$$\text{if } \mathbf{h} \in \Omega \text{ and } |\mathbf{h}| < \delta \quad \Rightarrow \quad |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| < \varepsilon.$$

Fix $\varepsilon > 0$.
Since f is differentiable at $\mathbf{a}$, there exists a matrix $J \in M_{m\times n}$ such that

$$\lim_{\mathbf{h}\to 0} \frac{|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - J\mathbf{h}|}{|\mathbf{h}|} = 0.$$

Take $\varepsilon' = 1$, $\exists \triangle$ such that if $0 < |\mathbf{h}| < \triangle$

$$\frac{|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - J\mathbf{h}|}{|\mathbf{h}|} < 1 \Rightarrow |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - J\mathbf{h}| < |\mathbf{h}|$$

$$\Rightarrow |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| = |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - J\mathbf{h} + J\mathbf{h}| \leq |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - J\mathbf{h}| + |J\mathbf{h}|.$$

Since $\exists M \geq 0$ such that $|J\mathbf{h}| \leq M|\mathbf{h}|$, this can be written as

$$|f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| < (1+M)|\mathbf{h}|.$$

Taking $\delta = \min\left(\triangle, \frac{\varepsilon}{1+M}\right)$, it follows that

$$0 < |\mathbf{h}| < \delta \Rightarrow |f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})| < (1+M)\frac{\varepsilon}{1+M} = \varepsilon.$$

But this means

$$\lim_{\mathbf{h}\to 0} (f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a})) = 0 \Rightarrow \lim_{\mathbf{h}\to 0} f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}).$$

□

Definition 4.2.5

Let $\Omega \subseteq \mathbb{R}^n$ be open, and consider the function $f : \Omega \to \mathbb{R}^m$ with components

$$f = (f_1, f_2, \ldots, f_m)^T.$$

Let $\mathbf{x} \in \Omega$, and let $i = 1, \ldots, m$ and $j = 1, \ldots, n$.
*The **partial derivative** of f_i with respect to x_j is*

$$\frac{\partial f_i}{\partial x_j}(\mathbf{x}) \equiv \lim_{t\to 0} \frac{f_i(\mathbf{x}+t\mathbf{e}_j) - f_i(\mathbf{x})}{t},$$

provided the limit exists.

Writing $f_i(x_1, \ldots, x_n)$ in place of $f_i(\mathbf{x})$ we see that $\frac{\partial f_i}{\partial x_j}$ is the derivative of f_i with respect to x_j, keeping the other variables fixed.

Definition 4.2.6

Let f be as defined above.
If all partial derivatives

$$\frac{\partial f_i}{\partial x_j} \equiv \frac{\partial f_i}{\partial x_j}(\mathbf{x}) \text{ exist } (i = 1, \ldots, m \text{ and } j = 1, \ldots, n),$$

*the **Jacobian matrix** of f is*

$$J_f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}.$$

Definition 4.2.7

*The determinant of the Jacobian matrix is called the **Jacobian** of f.*

Example.

> *Consider* $f : \mathbb{R}^2 \to \mathbb{R}^2$ *such that* $f(x, y) = \begin{pmatrix} x^2 + xy \\ x + y^3 \end{pmatrix}$.

$$\frac{\partial f_1}{\partial x} = 2x + y, \quad \frac{\partial f_1}{\partial y} = x$$
$$\frac{\partial f_2}{\partial x} = 1, \quad \frac{\partial f_2}{\partial y} = 3y^2$$

Therefore,

$$J_f(\mathbf{x}) = \begin{pmatrix} 2x + y & x \\ 1 & 3y^2 \end{pmatrix}$$

$$\det(J_f(\mathbf{x})) = (2x + y)(3y^2) - x(1) = 3y^2(2x + y) - x = 3y^3 + 6xy^2 - x.$$

Theorem 4.2.8

Let $\Omega \subseteq \mathbb{R}^n$ *be open, and suppose function* $f : \Omega \to \mathbb{R}^m$ *is* ***differentiable*** *at* $\underline{a}$. *Then* $f'(\underline{a}) = J_f(\underline{a})$.

In other words, the derivative of f *at* $\underline{a}$ *is the Jacobian matrix of* f *at* $\underline{a}$

Proof. Since f is differentiable at $\underline{a}$, there exists a matrix $J = f'(\underline{a})$ and a function R_a such that if $\underline{a} + \underline{h} \in \Omega$,

$$f(\underline{a} + \underline{h}) = f(\underline{a}) + J\underline{h} + R_a(\underline{h})|\underline{h}|, \quad \text{with } \lim_{\underline{h} \to 0} |R_{\underline{a}}(\underline{h})| = 0.$$

Without loss of generality, set $R_{\underline{a}}(\underline{0}) = 0$, so that $R_{\underline{a}}$ is continuous at $\underline{0}$. If we take $\underline{h} = t\underline{e_j}$, we seek to show that J is precisely the matrix of partial derivatives, $J_f(\underline{a})$.

$$f(\underline{a} + t\underline{e_j}) = f(\underline{a}) + tJ\underline{e_j} + R_{\underline{a}}(t\underline{e_j})|t\underline{e_j}|$$

Since $f_i = (f, \underline{e_i})$, then for $t \neq 0$,

$$\begin{aligned} f_i(\underline{a} + t\underline{e_j}) - f_i(\underline{a}) &= (f(\underline{a} + t\underline{e_j}) - f(\underline{a}), \underline{e_i}) \\ &= (tJ\underline{e_j} + R_{\underline{a}}(t\underline{e_j})|t|, \underline{e_i}) \\ &= t(J\underline{e_j}, \underline{e_i}) + (R_{\underline{a}}(t\underline{e_j}), \underline{e_i})|t| \end{aligned}$$

$$\Rightarrow \frac{f_i(\underline{a} + t\underline{e_j}) - f_i(\underline{a})}{t} = (J\underline{e_j}, \underline{e_i}) + (R_{\underline{a}}(t\underline{e_j}), \underline{e_i})\frac{|t|}{t}$$

Using the Cauchy–Schwarz inequality,

$$|(R_{\underline{a}}(t\underline{e_j}), \underline{e_i})| \leq |R_{\underline{a}}(t\underline{e_j})||\underline{e_i}| = |R_{\underline{a}}(t\underline{e_j})|$$

Since $\lim_{t \to 0} t\underline{e_j} = 0$ and $R_{\underline{a}}$ is continuous at $\underline{0}$,

$$\lim_{t \to 0} |R_{\underline{a}}(t\underline{e_j})| = |R_{\underline{a}}(0)| = 0 \Rightarrow \lim_{t \to 0} (R_{\underline{a}}(t\underline{e_j}), \underline{e_i})\frac{|t|}{t} = 0$$

Using the definition of the partial derivative,

$$\frac{\partial f_i}{\partial x_j}(\underline{a}) = \lim_{t \to 0} \frac{f_i(a + t\underline{e_j}) - f_i(\underline{a})}{t} = (J\underline{e_j}, \underline{e_i})$$

It remains to show that $(J\underline{e_j}, \underline{e_i})$ is the element in the i-th row and j-th column of matrix J.

$$Je_j = \begin{pmatrix} J_{11} & J_{12} & \cdots & J_{1n} \\ J_{21} & J_{22} & \cdots & J_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ J_{m1} & J_{m2} & \cdots & J_{mn} \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_{\text{row } j} = \begin{pmatrix} J_{1j} \\ J_{2j} \\ \vdots \\ J_{mj} \end{pmatrix}$$

So,

$$(J\underline{e_j}, \underline{e_i}) = (J_{1j}, \ldots, J_{mj}) \cdot (0, \ldots, 1, \ldots, 0) = J_{ij}$$

Hence,

$$\frac{\partial f_i}{\partial x_j}(\underline{a}) = J_{ij}.$$

□

*Existence of partial derivatives does **not** imply differentiability!*

Example.

> *Consider* $f : \mathbb{R}^2 \to \mathbb{R}$ *such that*
>
> $$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$
>
> Then
>
> $$f_x(0, 0) = \lim_{h \to 0} \frac{0}{h} = 0 = f_y(0, 0).$$
>
> *Show that* f *is however, not differentiable at* $\mathbf{0}$.

Proof. *f is not continuous at $(0, 0)$ and therefore cannot be differentiable by Theorem*
To show this, set $x = t$, $y = 0$

$$\lim_{t \to 0} f(t, 0) = \lim_{t \to 0} 0 = 0$$

Then set $x = y = t$

$$\lim_{t \to 0} f(t, t) = \lim_{t \to 0} \frac{t^2}{2t^2} = \frac{1}{2}$$

Hence, the limit depends on the path and does not exist.

□

Example.

> Can you show from the definition of differentiability that it is not differentiable?

Proof. Since both partial derivatives are 0, the 1×2 Jacobian matrix is:

$$J_f = (0, 0)$$

If f was differentiable at $(0, 0)$, we would have

$$\lim_{\mathbf{h} \to 0} \frac{|f(\mathbf{h} + \mathbf{0}) - f(\mathbf{0}) - J_f \mathbf{h}|}{|\mathbf{h}|} = 0.$$

Taking $\mathbf{h} = (h_1, h_2)$,

$$\frac{f(\mathbf{h} + \mathbf{0}) - f(\mathbf{0}) - J_f(\mathbf{h})}{|\mathbf{h}|} = \frac{\left| \frac{h_1 h_2}{h_1^2 + h_2^2} - 0 - (0, 0) \cdot (h_1, h_2) \right|}{(h_1^2 + h_2^2)^{1/2}} = \frac{|h_1 h_2|}{(h_1^2 + h_2^2)^{3/2}}.$$

To show this does not go to 0, take $h_1 = h_2 = t$:

$$\frac{|h_1 h_2|}{(h_1^2 + h_2^2)^{3/2}} = \frac{t^2}{(2t^2)^{3/2}} = \frac{1}{2^{3/2}} \cdot \frac{1}{t}.$$

But this does not go to 0 as $t \to 0$. □

Theorem 4.2.9

Let $\Omega \subseteq \mathbb{R}^n$ be open, and suppose function $f : \Omega \to \mathbb{R}^m$ has partial derivatives

$$\frac{\partial f_i}{\partial x_j} \text{ in an open ball around } \mathbf{a}.$$

Suppose these $\frac{\partial f_i}{\partial x_j}$ are continuous at $\mathbf{a}$.
Then f is differentiable at $\mathbf{a}$.

The proof is omitted (see Spivak "Calculus on Manifolds" for a proof). This is absolutely something we could do, but not in the time left!

Proof. It suffices to consider the case $m = 1$, so that $f : \mathbb{R}^n \to \mathbb{R}$. Then

$$\begin{aligned} f(a+h) - f(a) = {} & f(a^1 + h^1, a^2, \ldots, a^n) - f(a^1, \ldots, a^n) \\ & + f(a^1 + h^1, a^2 + h^2, a^3, \ldots, a^n) - f(a^1 + h^1, a^2, \ldots, a^n) \\ & + \cdots \\ & + f(a^1 + h^1, \ldots, a^n + h^n) - f(a^1 + h^1, \ldots, a^{n-1} + h^{n-1}, a^n). \end{aligned}$$

Recall that $D_i f$ is the derivative of the function g defined by $g(x) = f(x, a^2, \ldots, a^n)$. Applying the mean-value theorem to g we obtain

$$f(a^1 + h^1, a^2, \ldots, a^n) - f(a^1, \ldots, a^n) = h^1 \cdot D_1 f(b_1, a^2, \ldots, a^n)$$

for some b_1 between a^1 and $a^1 + h^1$.

Similarly, the ith term in the sum equals

$$h^i \cdot D_i f(a^1 + h^1, \ldots, a^{i-1} + h^{i-1}, b_i, \ldots, a^n) = h^i D_i f(c_i),$$

for some c_i. Then

$$\lim_{h \to 0} \frac{\left| f(a+h) - f(a) - \sum_{i=1}^n D_i f(a) \cdot h^i \right|}{|h|} = \lim_{h \to 0} \frac{\left| \sum_{i=1}^n [D_i f(c_i) - D_i f(a)] \cdot h^i \right|}{|h|}.$$

$$\leq \lim_{h \to 0} \sum_{i=1}^n |D_i f(c_i) - D_i f(a)| \cdot \frac{|h^i|}{|h|} \leq \lim_{h \to 0} \sum_{i=1}^n |D_i f(c_i) - D_i f(a)| = 0,$$

since $D_i f$ is continuous at a. □

Theorem 4.2.10: Chain Rule

Let $U \subseteq \mathbb{R}^n$, $V \subseteq \mathbb{R}^m$ both be open and let $g : U \to V$ be differentiable at $a \in U$. Let $f : V \to \mathbb{R}^p$ be differentiable at $g(a)$. *Then $F = (f \circ g)$ is differentiable at a and*

$$F'(a) = f'(g(a))g'(a).$$

Proof. For ease of notation, let $g(a) = b$. Since g is differentiable at a and f is differentiable at b, there exist Jacobian matrices $J_g = g'(a)$ and $J_f = f'(b)$ such that whenever $a + h \in U$, $b + z \in V$,

$$g(a+h) = g(a) + J_g h + R_{g,a}(h)|h| \tag{1}$$
$$f(b+z) = f(b) + J_f z + R_{f,b}(z)|z| \tag{2}$$

and

$$\lim_{h\to 0} |R_{g,a}(h)| = 0 \tag{3}$$

$$\lim_{z\to 0} |R_{f,b}(z)| = 0. \tag{4}$$

Without loss of generality, we set

$$R_{g,a}(0) = 0, \quad R_{f,b}(0) = 0,$$

so both functions are continuous.

Define

$$k : k(\mathbf{h}) = g(\mathbf{a}+\mathbf{h}) - g(\mathbf{a}) = J_g\mathbf{h} + R_{g,a}(\mathbf{h})|\mathbf{h}|. \tag{6}$$

Then

$$\begin{aligned} F(\mathbf{a}+\mathbf{h}) &= f(g(\mathbf{a}+\mathbf{h})) = f(g(\mathbf{a}) + k(\mathbf{h})) \\ &= f(g(\mathbf{a})) + J_f k(\mathbf{h}) + R_{f,g(a)}(k(\mathbf{h}))|k(\mathbf{h})| \\ &= F(\mathbf{a}) + J_f(J_g\mathbf{h} + R_{g,a}(\mathbf{h})|\mathbf{h}|) + R_{f,b}(k(\mathbf{h}))|k(\mathbf{h})| \end{aligned}$$

$$\Rightarrow \frac{F(\mathbf{a}+\mathbf{h}) - F(\mathbf{a}) - J_f J_g\mathbf{h}}{|\mathbf{h}|} = J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h})) \cdot \frac{|k(\mathbf{h})|}{|\mathbf{h}|}.$$

It remains to show that

$$\lim_{\mathbf{h}\to 0} \left| J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h})) \cdot \frac{|k(\mathbf{h})|}{|\mathbf{h}|} \right| = 0.$$

Proceeding,

$$\begin{aligned} \left| J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h})) \cdot \frac{|k(\mathbf{h})|}{|\mathbf{h}|} \right| &= |J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h}))| \cdot \frac{|J_g\mathbf{h} + R_{g,a}(\mathbf{h})|\mathbf{h}||}{|\mathbf{h}|} \\ &\le |J_f R_{g,a}(\mathbf{h})| + |R_{f,b}(k(\mathbf{h}))| \cdot \left| \frac{J_g\mathbf{h}}{|\mathbf{h}|} \right| + |R_{f,b}(k(\mathbf{h}))| \cdot |R_{g,a}(\mathbf{h})|. \end{aligned}$$

From Lemma, we know there exist $M_g, M_f \ge 0$ such that

$$|J_g\mathbf{z}| \le M_g|\mathbf{z}|, \quad |J_f\mathbf{z}| \le M_f|\mathbf{z}|.$$

Thus,

$$\left| J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h})) \cdot \frac{|k(\mathbf{h})|}{|\mathbf{h}|} \right| \le M_f|R_{g,a}(\mathbf{h})| + |R_{f,b}(k(\mathbf{h}))|M_g + |R_{f,b}(k(\mathbf{h}))||R_{g,a}(\mathbf{h})|.$$

We'll show that each of these terms go to 0 as $\mathbf{h}$ does.
From equation (4), we have

$$\lim_{\mathbf{h}\to 0} M_f|R_{g,a}(\mathbf{h})| = 0.$$

Since g is differentiable at $\mathbf{a}$, it is also continuous there (Theorem 2). Hence,

$$\lim_{\mathbf{h}\to 0} k(\mathbf{h}) = \lim_{\mathbf{h}\to 0} (g(\mathbf{a}+\mathbf{h}) - g(\mathbf{a})) = 0.$$

But since $R_{f,b}$ is continuous by construction,

$$\lim_{\mathbf{h}\to 0} R_{f,b}(k(\mathbf{h})) = R_{f,b}\left(\lim_{\mathbf{h}\to 0} k(\mathbf{h})\right) = R_{f,b}(0) = 0.$$

Hence,

$$\lim_{\mathbf{h}\to 0} |R_{f,b}(k(\mathbf{h}))|M_g = 0.$$

By the product of limits,

$$\lim_{\mathbf{h}\to 0} |R_{f,b}(k(\mathbf{h}))R_{g,a}(\mathbf{h})| = 0.$$

Finally then, by the squeeze theorem, we have our result, and

$$\lim_{\mathbf{h}\to 0}\left|J_f R_{g,a}(\mathbf{h}) + R_{f,b}(k(\mathbf{h}))\cdot\frac{|k(\mathbf{h})|}{|\mathbf{h}|}\right| = 0,$$

as required. □

Definition 4.2.11

Let $U \subseteq \mathbb{R}^n$. We say $f : U \to \mathbb{R}^m$ is ***continuously differentiable****, if f is differentiable in U and all its partial derivatives are continuous in U.*

Chapter 5

Inverse function theorem

5.1 In one dimension

Lemma 5.1.1

If f is a continuous and one-to-one function defined on an interval and $f'(x_0) = 0$, then f^{-1} is not differentiable at $f(x_0)$.

Proof. We have $f^{-1}(f(x)) = x$. If f^{-1} was differentiable at $f(x_0)$, then the chain rule gives $(f^{-1})'(f(x_0))f'(x_0) = 1$. Hence $(f^{-1})'(f(x_0)) \cdot 0 = 1$. Contradiction. □

Example.

> *Consider $f : f(x) = x^3$.*

Proof. Since $f'(0) = 0$, the function is not differentiable at 0. □

Lemma 5.1.2

Let $I \subset \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be continuous and strictly monotone. Then $f(I)$ is an interval.

Proof. Assume f is strictly increasing. Let $y_1, y_2 \in f(I)$. W.L.O.G, assume $y_1 \neq y_2$ and $y_1 < y_2$. Then $\exists x_1 < x_2$: $f(x_1) = y_1, f(x_2) = y_2$.

Pick any y: $f(x_1) < y < f(x_2)$. By IVT, $\exists x : x_1 < x < x_2$, and $f(x) = y$. Hence, $y \in f(I)$. This means $f(I)$ is an interval. □

Lemma 5.1.3

Let $I \subset R$ be an interval, and let $f : I \to R$ be strictly monotone. Then f is invertible on the range of f, and $f^{-1} : f(I) \to I$ is strictly monotone and continuous

Proof. Assume f is strictly increasing. We first prove invertible. By definition, f is surjective on $f(I)$, so we just need to show one-to-one. Let $s_1 \neq s_2 \in I$. Then w.l.o.g, assume $s_1 < s_2$. Since f is strictly increasing, we have $f(s_1) < f(s_2)$. Thus, $f(s_1) \neq f(s_2)$.

Now suppose $t_1, t_2 \in f(I)$, with $t_1 < t_2$. If $f^{-1}(t_1) \geq f^{-1}(t_2)$, this would imply $f(f^{-1}(t_1)) \geq f(f^{-1}(t_2)) \Rightarrow t_1 \geq t_2$. Contradiction. Hence, f^{-1} is strictly increasing.

We now need to show f^{-1} is contiuous. We'll show this is true for an open interval. Let $I = (a, b)$ with $a < b$. Let $y_0 \in f(I)$ and $x_0 = f^{-1}(y_0)$. WTS $\forall \epsilon > 0, \exists \delta > 0$ such that if $|t - y_0| < \delta, t \in f(I) \Rightarrow |f^{-1}(t) - f^{-1}(y_0)| < \epsilon$

Fix $\epsilon > 0$ and define $\epsilon' = \min\{\epsilon, \frac{b-x_0}{2}, \frac{x_0-a}{2}\}$.

$$x_0 < x_0 + \epsilon' \leq x_0 + \frac{b - x_0}{2} = \frac{x_0 + b}{2} < b$$

Similarly $x_0 > x_0 - \epsilon' > a$. So $x_0 + \epsilon', x_0 - \epsilon' \in I, f(x_0 + \epsilon'), f(x_0 - \epsilon')$ exists.

Define $\delta_1 = f(x_0 + \epsilon') - f(x_0) > 0$ and $\delta_2 = f(x_0) - f(x_0 - \epsilon') > 0$. Take $\delta = min\{\delta_1, \delta_2\}$. Suppose $t \in f(I), |t - y_0| < \delta$. Then

$$-\delta_2 \leq -\delta < t - y_0 < \delta \leq \delta_1$$

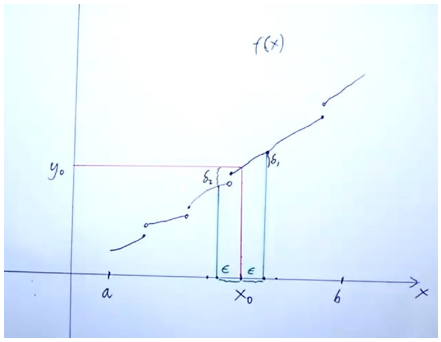
$$f(x_0 - \epsilon') = y_0 - \delta_2 < t < y_0 + \delta_1 = f(x_0 + \epsilon')$$

Since f^{-1} is strictly monotone,

$$x_0 - \epsilon \leq x_0 - \epsilon' < f^{-1}(t) < x_0 + \epsilon' \leq x_0 + \epsilon$$

$$\Rightarrow |f^{-1}(t) - f^{-1}(y_0)| < \epsilon$$

□



Lemma 5.1.4

Let $I, J \subset \mathbb{R}$ be intervals. Let $f : I \to J$ be strictly monotone, surjective, and differentiable at $x \in I$, with $f'(x) \neq 0$. Then the inverse f^{-1} is differentiable at $y = f(x)$ and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} = \frac{1}{f'(x)}.$$

Proof. By lemma 3, f has a continuous inverse. Since J is an interval, there are points arbitrarily close to y from both sides, and f^{-1} is defined by points in an open interval around y.

We want to show that

$$\lim_{k \to 0} \frac{f^{-1}(y + k) - f^{-1}(y)}{k} = \frac{1}{f'(x)}$$

Define $h(k) = f^{-1}(y + k) - f^{-1}(y), \forall k : f^{-1}(y + k)$ exists. Since f^{-1} is continuous, h is also continuous. Since f^{-1} is strictly monotone, we know $h(k) = 0$.

$$f^{-1}(y) + h(k) = f^{-1}(y + k)$$

$$\Rightarrow f(x + h(k)) = y + k$$

$$\Rightarrow k = f(x + h(k)) - f(x)$$

Now using Q15 of P5,

$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} = \lim_{k \to 0} \frac{f(x + h(k)) - f(x)}{h(k)}$$

$$\Rightarrow \lim_{k \to 0} \frac{f^{-1}(y + k) - f^{-1}(y)}{k} = \lim_{k \to 0} \frac{h(k)}{f(x + h(k)) - f(x)}$$

$$= \lim_{k\to 0} \frac{1}{\frac{f(x+h(k))-f(x)}{h(k)}} = \frac{1}{\lim_{k\to 0} \frac{f(x+h(k))-f(x)}{h(k)}} = \frac{1}{f'(x)}$$

□

Theorem 5.1.5: Inverse function theorem

Let $f : (a, b) \to \mathbb{R}$ be a continuously differentiable function, where $a < b$.
Suppose $x_0 \in (a, b)$ and $f'(x_0) \neq 0$. Then there exists an open interval $I \ni x_0$ such that $f|_I$ is invertible. Moreover, if $J = f(I)$, then $f^{-1} : J \to I$ is differentiable, and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad \forall y \in J.$$

Note 1: *Continuously differentiable* means that f' is continuous.
Note 2: $f|_I$ means the restriction of the domain of f to I.

Proof. Suppose $f'(x_0) > 0$ (similar if < 0).
As f' is continuous, $f'(x) > 0, \forall x \in$ some interval I. This means f strictly increasing on I.
As f is differentiable, it is continuous. So by lemma, J is an interval. Hence f is 1-1 and differentiable on I. Hence, lemma applies. □

Conclusion: In single variable calculus, if f is continuously differentiable, and $f'(x_0) \neq 0$, then f is locally invertible near x_0 and

$$(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}.$$

The derivative of the inverse is the inverse of the derivative.

5.2 In multiple dimensions

We are now in a position to consider a multivariable generalisation of the inverse function theorem:

Lemma 5.2.1

*If $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is a **continuous and invertible function** and*

$$\det(J_f(\mathbf{x}_0)) = 0,$$

*then f^{-1} is **not** differentiable at $f(\mathbf{x}_0)$.*

Proof. We have

$$f^{-1}(f(\mathbf{x})) = \mathbf{x} = I\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n.$$

If f^{-1} <u>were</u> differentiable at $\mathbf{x}_0$, the multivariable chain rule would apply. Take the derivative of both sides:

$$J_{f^{-1}}(f(\mathbf{x}_0)) \cdot J_f(\mathbf{x}_0) = I \quad \Rightarrow \quad J_{f^{-1}}(f(\mathbf{x}_0)) = (J_f(\mathbf{x}_0))^{-1}$$

But if $\det(J_f(\mathbf{x}_0)) = 0$, this matrix has no inverse! Contradiction.

□

Theorem 5.2.2: Inverse Function Theorem (Multivariable version I)

Consider $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$. Suppose f is continuously differentiable at $\mathbf{x}_0$ and $f'(\mathbf{x}_0) = J_f(\mathbf{x}_0)$ is an invertible matrix. Then there exist open sets $U \ni \mathbf{x}_0$ and $V \ni f(\mathbf{x}_0)$ such that $f|_U : U \to V$ is invertible. Moreover f^{-1} is differentiable at $f(\mathbf{x}_0)$ and

$$J_{f^{-1}}(f(\mathbf{x}_0)) = (J_f(\mathbf{x}_0))^{-1}.$$

In other words, the derivative of the inverse is the matrix inverse of the derivative.
Upshot: suppose J_f is cts and invertible at $\underline{x}$. Then f is locally invertible and the above theorem applies.

Proof. Omitted. □

Theorem 5.2.3: Inverse Function Theorem (Multivariable version II)

Let $\Omega \subseteq \mathbb{R}^n$ be open, and consider $f : \Omega \to \mathbb{R}^n$. Suppose f is continuously differentiable ***in an open set containing*** $\mathbf{x}_0$ *and $f'(\mathbf{x}_0) = J_f(\mathbf{x}_0)$ is an invertible matrix. Then there exist open sets $U \ni \mathbf{x}_0$ and $V \ni f(\mathbf{x}_0)$ such that $f|_U : U \to V$ has a* ***continuous inverse****. Moreover $f^{-1} : V \to U$ is differentiable and satisfies*

$$J_{f^{-1}}(\mathbf{y}) = \left(J_f(f^{-1}(\mathbf{y}))\right)^{-1} \quad \forall \mathbf{y} \in V.$$

This theorem is more powerful.

Proof. Omitted. □

Example.

(Final 2014). *Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be given by*

$$f(x, y) = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}.$$

1. Calculate the Jacobian matrix of f and its determinant.
2. Prove that f is locally invertible at every point in $\mathbb{R}^2$.
3. Is f globally invertible?

1. We have
$$J_f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$$
$$\det(J_f(\mathbf{x})) = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x}.$$
2. $\det(J_f(\mathbf{x})) \neq 0$ at every point in $\mathbb{R}^2$.
   J_f is continuous $\Rightarrow$ f is continuously differentiable.
   Thus, the inverse function theorem implies that f is locally invertible at every point.
3. f cannot be globally invertible as it is not one-to-one:
$$f(0, 0) = f(0, 2\pi) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Example.

(Final 2015). Let $f(x, y) = \begin{pmatrix} x^2 y + yx \\ 2xy + xy^2 \end{pmatrix}$.

Show that f in an open set containing $(1, 1)^T$ has an inverse f^{-1} defined in an open set containing $(2, 3)^T$. Find the derivative of f^{-1} at $(2, 3)^T$.

Note $f(1, 1) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

$$J_f(x, y) = \begin{pmatrix} 2xy + y & x^2 + x \\ 2y + y^2 & 2x + 2xy \end{pmatrix}$$

$$J_f(1,1) = \begin{pmatrix} 3 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\det(J_f(1,1)) = 3 \cdot 4 - 2 \cdot 3 = 6 \neq 0.$$

Hence, by the inverse function theorem, f is locally invertible.

$$J_{f^{-1}}(2,3) = \left(\begin{pmatrix} 3 & 2 \\ 3 & 4 \end{pmatrix} \right)^{-1} = \frac{1}{6} \begin{pmatrix} 4 & -2 \\ -3 & 3 \end{pmatrix}$$